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BES REVIEW

Quarterly Journal of Broadcast Engineering Society (India)



Glimpses of BES Expo 2025

















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BES Review

October'25 - December'25

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Published by :
Aftab Ahmad, *Hon. Secretary,*
on behalf of
Broadcast Engineering Society (India)
912, Surya Kiran Building,
19, K.G. Marg, New Delhi-110001
Tel. : + 91-11- 23316709
E-mail : bes@besindia.com

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Pragati Creations
486/1, Dilshad Garden Ind. Area,
G.T. Road, Delhi-110095
e-mail: pragaticreations5@gmail.com
Ph.: 011-49405140, 9312438440

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Editorial



Dear Friends,

Welcome to this edition of the BES Review, which is being published after a break of almost three years. As we step into 2026, the key discussion has moved towards a more mature and strategic development of Hybrid Broadcasting. The industry has entered a phase of calculated consolidation, where traditional & trusted reach and personalized digital streaming are no longer rivals, but partners in a new operating baseline. Aging SDI equipment, high maintenance costs, and the need for flexible, software-defined solutions are pushing broadcasters to adopt cloud-native, IP-driven infrastructure. Artificial Intelligence has leapfrogged from being a "technical experiment" to an operational necessity. The greatest opportunity lies in combining the "human touch" with the productivity of AI.

It is great feeling that annual event BES EXPO which is being held in New Delhi during 29-30-31 January, 2026. We will have hundreds of Exhibiters coming from various parts of the world to showcase the latest broadcast technologies and equipment at BES EXPO-2026 and Technology experts talking on state-of-the-art technologies. The biggest challenge of broadcasting is not just technology adoption, but the widened skills gap between traditional broadcast engineering and IT expertise.

We have tried to present a wide spectrum of articles in this BES Review that includes 'Augmented Learning's for Repair of R&S Make TV Transmitters' by Mr M. S. Duhan;

'An Overview of the Third Generation AVS Video Coding Standard (AVS3)' by Ms Zhou Yun; 'Significance of Frequency Band for Success of Digital Radio Broadcasting' by Mr P. DAS; and 'SVURG/Z Mod – An Introduction' by Mr Satyapal. Inside you will also find our regular features "Local Chapter Report" and "News from Broadcasting World".

I would like to extend my sincere thanks to all the authors who have contributed for this edition. I would like to place on record my special thanks to ABU and especially to Mr Nadeem, Secretary-General and Mr Veysel Binbay, Director, Technology & Innovation for arranging international article for this edition of BES Review.

Best wishes and looking forward to meet many of you at BES EXPO -2026.

Rajeev

Rajeev Kumar
rajeev00kumar@yahoo.com

From the President



Dear Hon'ble Members,

I am highly delighted in reaching out to you to inform, BES review is being published againafter abreak. The editor and his team have brought out a number of interesting articles from great authors for the benefit of Members.

This time, BES isorganizing the30thInternational Conference and Exhibition on Broadcast Media Technology from 29-31 January, 2026 in Hall No 12A, Bharat Mandapam, New Delhi with ever-increasing pace and energy.This year, we gather under the theme 'Broadcast Intelligence Innovation: Make in India for the World,' a vision that moves beyond simple infrastructure toward a future defined by intelligent systems and technological self-reliance.BES EXPO 2026 brings the broadcast & media community together with a focus on"Make in India" initiatives in broadcasting, Broadcast intelligence, integrated workflow and the impact of AI on media and broadcasting.

Our industry has reached a pivotal juncture where the boundaries between broadcast, broadband, and telecommunications have effectively dissolved, creating a singular, converged ecosystem that demands both localized expertise and global scalability. The current television landscape is defined by a "dual ecosystem" where traditional linear broadcasting and digital-first platforms have fully converged. The industry is shifting from hardware-heavy infrastructures to IP-native workflows and Cloud-based production, allowing for unprecedented scalability and cost efficiency. The emergence of Direct-to-Mobile (D2M) technology and the rise of Connected TV have redefined the "living room experience," turning TV into a personalized, cross-screen medium that integrates high-quality linear content with interactive digital services.

Right from being a source of information and keeping up with the day-to-day news, to now turning into a complete entertainment package, this medium has played a huge role in the country's communication and information industry.The creation of multiple touchpoints for the audience has led to an expansion in customer engagement. Associating with the audio platforms sparked the idea of the perfect amalgamation of radio and digital which gave rise to a host of opportunities to distribute content to a global audience.

Dear Readers, Broadcasting is changing tremendously much beyond our imagination. The way content is created, distributed and consumed has transformed the broadcast landscape. The young generation with the vast broadband usage is consuming the audio and video content more than the traditional broadcasting. OTT platforms and Social media applications are disrupting the broadcasting industry. It has created lots of opportunities for broadcasters and hence should be considered as an opportunity than threat.

BES EXPO 2026 will bring together industry professionals, exhibitors, and policymakers to discuss the future of media technology.I am sure all delegates and members will get benefitted by attending it.

With best wishes!

Sunil

BES (I) Council 2024-2026

Name	Designation	Mobile	E-mail
Sunil	*President*	9868230260	sunil.ddi@gmail.com
Mukesh Kumar	*Immediate Past President*	9711198244	mukeshkumar1965@gmail.com
C.B.S. Maurya	*Past President*	9650296668	cbsmaurya@gmail.com
Aftab Ahmad	*Hon. Secretary*	9899016159	aftabahmad1@rediffmail.com
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K.K. Rao	*Vice President*	9868732384	krishnarao73@yahoo.co.in
Devesh Kumar Sachan	*Vice President*	9811825982	Dksachan08@gmail.com
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C.K. Jain	*Council Member*	9810104938	ckjaindd@gmail.com
Sandeep Srivastava	*Council Member*	8351990180	sandeep@prasarbharati.gov.in
P. Das	*Council Member*	9873293421	pdasdelhi@gmail.com
Rajeev Kumar	*Council Member*	9868366232	rajeev00kumar@yahoo.com
Shailesh Kumar Kanauje	*Council Member*	9871290242	shaileshkanauje@rediffmail.com
K. Murugan	*Council Member*	9971159629	murugan_dde@rediffmail.com
Rajesh Kumar Sharma	*Council Member*	8076133602	rksharma@prasarbharati.gov.in
Abhishek Agrawal	*Council Member*	9868539572	abhiagra31@gmail.com
M.P. Singh	*Council Member*	9999200346	mpsinghwin@rediffmail.com
N. Namassivaya	*Co-opted Member*	9866692214	nnamassivaya@gmail.com
Jagdish W Bakal	*Co-opted Member*	9820723812	bakaljw@gmail.com

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8.	**Local Chapter Coordination Committee**	Shri Anil Kumar Mangalgi	Shri C.K. Jain & Shri N.V. Ramanna

Report on Objectives and Activities of Broadcast Engineering Society (India)

Manoj Gupta, *Honorary Treasurer*

Introduction

The Broadcast Engineering Society (India), commonly abbreviated as BES (I), is the premier professional society in India representing individuals and organizations engaged in broadcasting, media, and allied technologies. Founded in 1987, BES (India) functions as a non-profit, non-governmental, and voluntary organization dedicated to advancing the field of broadcast and media engineering through research, knowledge sharing, training, and collaborative initiatives.



BES (India) serves as a platform for engineers, technocrats, media professionals, academicians, and equipment manufacturers to interact, exchange ideas, and stay abreast of technological developments in the field of broadcasting.

Vision and Mission

Vision: To be the national leader in promoting technological advancement and knowledge sharing in broadcasting and media technology.

Mission: To support and strengthen the professional development of broadcasting engineers and to foster innovation, standardization, and skill-building through knowledge dissemination and industry engagement.

Objectives of BES (India)

- Promote Technological Excellence – To create a professional environment that encourages the pursuit of excellence in broadcast engineering.
- Facilitate Knowledge Exchange – To act as a hub for the exchange of ideas, case studies, and best practices.
- Encourage Research and Development – To promote R&D in broadcasting and media technologies.
- Disseminate Information on Emerging Technologies – To educate stakeholders on developments in key broadcasting areas.
- Foster Industry–Academia Collaboration – To bridge the gap between academia and industry.
- Support Policy and Standardization Efforts – To act as a knowledge partner for government and regulatory bodies.
- Capacity Building and Training – To enhance the skill sets of professionals and students.

Major Activities of BES (India)

i) BES EXPO – Annual International Conference & Exhibition

The BES EXPO, held annually in New Delhi, is South Asia's largest gathering of broadcast professionals. It includes: - Exhibition: Participation from global and Indian technology providers. - Conference: Technical sessions, keynote speeches, and panel discussions. -



Audience: Engineers from AIR, Doordarshan, private broadcasters, and system integrators.

ii) Technical Seminars, Webinars and Workshops

Organized across the country focusing on topics like Digital Radio, IP broadcasting, OTT, cloud playout, UHD/4K workflows, and more.

iii) Publication: BES Review (Technical Journal)

A quarterly journal featuring peer-reviewed articles, product reviews, and regulatory updates. It is circulated to members and available online.

iv) Student and Young Engineer Development

Initiatives include student memberships and to encourage youth participation in broadcast technology.

v) Collaboration with Global Bodies

BES (India) collaborates with international bodies like ABU, AIBD, IEEE EBU, and WBU-TC to bring global standards and practices to India.

vi) Institutional and Corporate Memberships

Memberships are open to individuals and institutions with benefits like technical events, networking, and exhibition participation.

Future Plans and Focus Areas

- Promoting Green Broadcasting and energy-efficient technologies.
- Supporting D2M Broadcast Trials and NextGen TV rollout.
- Expanding regional chapters for wider engagement.
- Enhancing digital content delivery and OTT readiness.
- Strengthening India's capabilities in disaster communication using broadcast networks.

BES Contribution in WAVES 2025 at Mumbai

BES (India) made a significant contribution to the success of the WAVES 2025 event held in Mumbai. WAVES 2025, a prestigious industry event focusing on media, entertainment, and broadcast technologies, provided a platform for technology showcase, knowledge exchange, and strategic discussions among key stakeholders in the media and broadcast sector.

BES participated as a key knowledge partner, Curated and hosted a dedicated technical session on 'Next-Generation Broadcast Technologies' featuring speakers from broadcasters, global leaders and technocrats.

The participation of BES (India) was well-received and reinforced its leadership in the Indian broadcast technology domain. It also opened new avenues for collaboration with industry bodies, academia, and startups focused on next-gen media solutions.

Conclusion

The Broadcast Engineering Society (India) plays a vital role in shaping the broadcasting and media technology landscape in India. Through events like BES EXPO, professional training, and technical publications, it continues to empower professionals and contribute to the modernization of Indian broadcasting. With a focus on innovation, collaboration, and capacity building, BES (India) remains a cornerstone institution for broadcast engineers in the country.





Augmented Learning's for Repair of R & S make TV Transmitters

M. S Duhan

1. Introduction:

In the DD Network, there are number of Rohde & Swartz(R&S) makeliquid-cooled VHF &UHF transmitters for analog TV High Power Transmitters. The rated power ranges from 5 kW to 20 kW. Major organs of Transmitters are;Exciter {*Encoder&Digital pre-corrector, I/Q Modulator, Synthesizer, Central Control Unit(CCU), Control unit, Motherboard, Power supply}*, Power Amplifiers, Splitters and Combiners, Diplexer, RF Rectifier, Rack controllers, Heat Exchanger, Pump & PUC etc. A series of workshop have been organized by DD in past yearsby resource persons from field units and O/o ADG(NZ). This paper is coiled around various fault analysis and repairs carried out on defective modules received from all over DD Network during such workshops. It will help in augmented learnings and capacity building of DD Engineers.

2. General Architecture:

R & S Transmitters have various protection/monitoring facilities likeabsence of video input signal, reflection monitoring (VSWR), amplifier and power supply monitoring, over temperature of power amplifier, liquid flow monitoring, inlet liquid temperature monitoring, AC voltage monitoring and AC phase monitoring. However, still the failures of modules/system do happen. During the workshops, the itinerary was chalked encompassing illustration of theory of technology and imparting hand-on training for fault diagnosis, demonstrations for replacement of defective semiconductors, firmware loadings and alignments etc.

2.1. The architecture of TV Transmitter: -

Analog TV transmitters for up to 10 kW power, including up to five video amplifiers and one audio amplifier, are accommodated in a single rack and 20 kW TV Transmitters have two racks with twelve video amplifiers and two audio amplifiers. Harmonic filter is used to restrict the out of band emissions. A directional coupler is used to decouple part of the output power for test purposes. For diagnostic purposes, two service connectors are provided for determining transistor currents, power supply voltages and currents as well as all parameters from the RF detectors. The values can be measured with a simple multi-meter, or more conveniently with the optional Test Box R&S GT610 A1 and displayed on a laptop.Fig.1. shows the general architecture of the TV Transmitter.

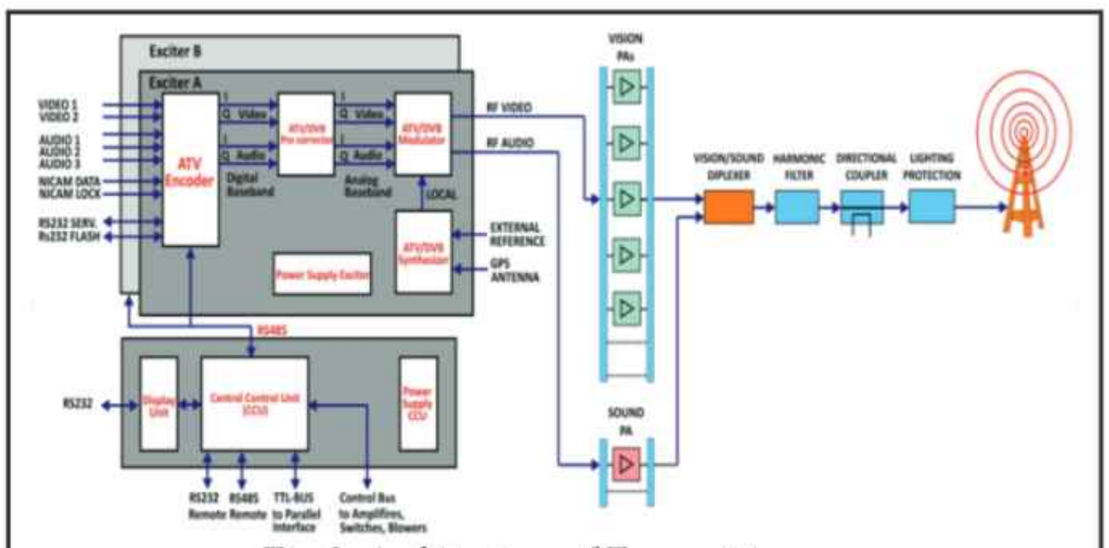


Fig.1. Architecture of Transmitter

The exciter performs signal conditioning for the RF signal to be transmitted. The CCVS input signal is modulated onto a carrier signal at a certain frequency (channel frequency). The amplification process in the output stages of the TV transmitter as well as the RF filter result in distortion in the actual output signal. However, this distortion must be eliminated from the signal that is transmitted, which is why pre-correctors are used in the exciters. The power filters at the transmitter output create linear phase and amplitude distortion on the filter edges that can reduce the signal/noise ratio. In case of linear distortion, the amplitude as well as the group delay for a specific frequency range is not constant. Looking at the analogue television signal, linear distortion results in a sort of slurring of the waveform.

2.2 Learning Theology :Technology and Repairs

2.2.1. ATV Encoder: This unit receives an analogue video signal & audio signals and digitizes them. The digital signal is processed and converted into two quadrature baseband signals with a Hilbert transform. Hilbert Transform of signal x(t) is defined as

Sr No	Component
1	Max 701
2	Max 3232EC
3	Max 708T
4	Max 3491
5	TLO 074

Table. 1. Critical component Failures in Encoders

transform of a signal in which phase of each component of signal is shifted by ± 90^0. During this procedure, the required SSB characteristic is also set. The encoder also contains a microcontroller which drives the whole exciter and handles communication with the CCU (central control unit). A program memory is provided as a peripheral for the micro controller.

The digital baseband signals in the video and audio paths are applied to the pre-corrector. The encoder's program memory stores all its software and firmware. It is flash memory which means that software can be updated without replacing any hardware.The encoder's program memory stores all its software and firmware. It is flash memory which means that software can be updated without replacing any hardware.The update is performed via the RS232 "FLASHUP" interface at the front of the encoder.Critical failures of components in the encoder are shown in fig.1.The learning experience in workshops include fault analysis and replacement of trans-receiver ICs, Transceiver Max 3232EC, Buffers & Line Drivers, high-speed JFET input single operational amplifier(TL074) , Max 701 (supervisory circuits used to monitor the power supplies in microprocessor (µP) and digital systems), andreloading of firmware's.While replacing the new encoder, establish a connection by connection a common RS-232 cable between COM1 or COM2 with PC and the transmitter interface on exciter from transmitter(Local Tx). Now the setting of the module can be changed. If the motherboard recognize the new unit, Switch ON isolating switch Q1 . Now transfer the data stored on PC from the old unit. Check the synthesizer frequency and switch on the Transmitter.

Procedure for updating software in new Encoder

a) Put the assembly in circuit in exciter rack say 'B'.
b) Select this exciter as reserve and make it on.
c) From PC/Lap Top open the software 127atv.
d) Connect CCU output with PC.
e) Press Reset MC.
f) Now press Flash Up and keep it pressed.
g) Press Reset MC and release.
g) Release Flash up.
h) Enter in software 127atv Flash.1bat in PC and it will take 6-7 minutes to load it.
i) Reboot from CCU.
j) Load the ATV Encoder parameter with GUI (sdb) by using menu> file> file manager>Load selected parameter (e.g. Exc. A)
k) Check frequency of synthesizer and TV Standard (B/G).
l) Test for smooth functioning of XTR.

2.2.2 Modulator &Equalizer Assembly: The amplifiers are normally operated for high efficiency which causes nonlinear signal distortion. Thus, the instantaneous gain and the phase shift of the signal are functions of amplitude. This creates spurious frequencies in the signal spectrum. Band limiting at the transmitter output causes linear phase distortions in the



Fig. 2. Modulator's Open View

vicinity of the filter skirts, which lead to colour changes or shadows in the case of ATV. This is where the ATV/DVB equalizer comes in. The ATV equalizer is made up of a group-delay equalizer and a linearity pre-

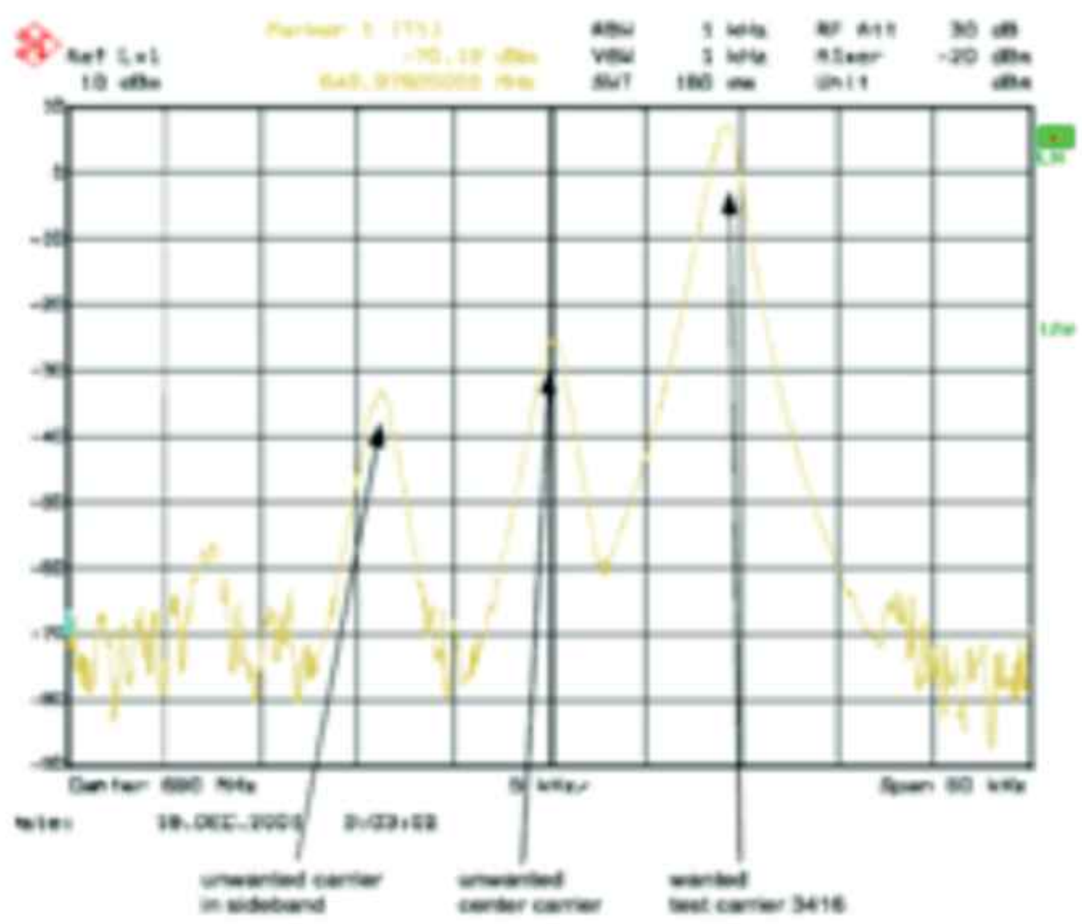


Fig.3. 1 –Bias and Q- Bias adjustments

Replacement of Modulator-Equalizer Assembly:

a) Put the assembly in circuit in exciter rack say 'B'.

b) Select above exciter as reserve and make it on.

c) Take video output of above Modulator and connect it to spectrum analyzer with the following settings: -

i) Centre Frequency: 471.25 MHz. (Vision Carrier)

ii) Span: 20 MHz. iii. Amplitude: -20 dBm.

iv) RBW: 10 KHz. v). VBW: 30 Hz.

d) From PC/Lap Top select Exc- RF- Video

e) Adjustment is done in standby Exciter or PA Off position.

f) Switch OFF the Regulation: EXC RF>RF Out Video Menu > Regulation > OFF.

g) Manually enter RF Level between 30 and 60 %.

h) Put SIDE SUPPR. "ON". The sound sub carrier is routed to vision modulator. {Eliminate centre carrier (vision frequency) and lower sideband carrier (-5.5 MHz) -50, -60 dB related to upper test carrier. Sound sub carrier is adjusted without additional upper test carrier.}

i). There may be unwanted sidebands besides wanted carrier frequency. Suppress the unwanted centre carrier to 50 dB below the wanted carrier by adjustment of I –Bias and Q- Bias adjustments *(Use the two control elements alternatively and reduce the step width as the suppression gets better)*. Suppress unwanted centre carrier in the left side band to at least 60 dB using Level and Phase adjustments. Suppress the unwanted centre carrier to 65 dB below the wanted carrier by adjustment of I –Bias and Q- Bias adjustments. In case we have identical frequency, we can copy the 'I-Bias, Q-Bias, Level% and Phase-Carr %' without performing above steps.

j) Now select-Exc-RF-Audio. Switch off Regulation and Sub carrier of Audio CH1 ON and Modulation Off. Connect audio output to spectrum analyzer and make adjustments as in step 'i'.

k) Modulator is now reading for operation.

corrector. Correction is performed at the digital baseband level. The equalization function itself is performed by FPGAs (field-programmable gate array). With the aid of the FIR (Finite Impulse Response) control FPGA, a 54 MHz system clock is derived from the internal clock by doubling for ATV. The system clock synchronizes signal transfer to the input buffers. The frequency response filter checks whether the instantaneous amplitude is above a specific threshold. If this is the case, the signal component to which the frequency response information is to be added is calculated using the current phase of the vector formed from the I and Q components. This signal component is fed through two complex FIRfilters, one of which is able to tilt the amplitude frequency response about the vision carrier (corresponds to the frequency 0 Hz in the baseband) using a setting parameter. The second filter tilts the phase frequency response about the vision carrier.The digital signals are then converted into analog baseband signals which are fed to the modulator. The ATV/DVB modulator generates the RF vision and sound carriers by direct quadrature modulation. The signals are filtered and amplified and then fed to the output stages.The RF SSB characteristic required is generated by complex filtering in service. The modulator contains two processing paths, one for the vision carrier and one for the sound carrier.In the combined version, the sound carrier path is dispensed with. An RF monitor is provided in the modulator. The details of general component failures in Modulator are shown in Table 2.

Fault	Component failure/Tests
Video Output is available but No Audio or Low Audio Output	Hybrid Amplifier N9 Failure.Pin No 3 or output port of IC having the voltage less than 4.5 Volt or higher than 5 V. If no voltage available at output pin, check the voltage at L20. If still there is no voltage and L20 is okay, that means IC is defective.
Video Output is OK and Audio Output having reflection	Check for RF Audio Output for side bands and adjust if required. Otherwise check IC D36, Transistors T1 and T2.
Audio Output is available but Video Output is not available	Check IC N53, D13 HELA 10 IC and IC D42. To test HELA 10 IC, remove L31 and L32, put a multi-meter in series in current mode. It should be 10 mA. Replace burnt components if any.

Table 2.0. Faults & Component Failures in Modulator

It outputs measurement data on the vision and sound carrier at the exciter output and ahead of the antenna. The exciter output signals are controlled and monitored in this way. The values measured at the antenna are used for monitoring but also for displaying the vision and sound carrier power.The two local signals are phase- and amplitude-stabilized before they are appliedto the mixer. After mixing, the two signal components are added in a combiner, giving the RF signal. The Graph shown in Fig 3 illustrate alignments of I-Bias and Q-Bias. The procedure for replacement of Modulator-Equalizer is also described in above block.

FaultComponent failureNo outputDue to 120 MHz G1 Crystal Failure or no output from modulatorNo Reference OutputDue to failure of 10 MHz B1 Crystal or IC D9Low OutputDue to Monolithic Amplifier MGA 81563 N15 Failure.Due to Monolithic Amplifier MGA 81563 N16 FailureNo Frequency ChangeDue to Trans-Receiver D12 MAX 232A Failure.Table 3.0. Faults and Failures in Synthesizer**2.2.3. Synthesizer:**

The ATV/DVB synthesizer generates the vision carrier frequency required for modulation. The synthesizer generates the vision carrier frequency required to generate vision and sound carriers in the case of quadrature modulation. To change the vision carrier frequency, the new frequency must be set and the I/Q modulators in the video and audio paths readjusted. The measurement of the output power using a power meter with a peak-power sensor and a spectrum analyzer and the adjustment of the quadrature modulator must be performed at the exciter outputs that are in service. The only check point while replacing the new synthesizer is to check the RF Frequency.*Output level should be about -35 dBm.*The critical failure of component is detailed in Table 3.0.

Fault	Component failure
No output	Due to 120 MHz G1 Crystal Failure or no output from modulator
No Reference Output	Due to failure of 10 MHz B1 Crystal or IC D9
Low Output	Due to Monolithic Amplifier MGA 81563 N15 Failure. Due to Monolithic Amplifier MGA 81563 N16 Failure
No Frequency Change	Due to Trans-Receiver D12 MAX 232A Failure.

Table 3.0. Faults and Failures in Synthesizer

2.2.4. Motherboard / Emergency operation: The motherboard interconnects the exciter modules and links the exciter to the CCU. The motherboard forms the interface to the transmitter rack (with cooling system, power supplies and amplifiers) for transmitter control. The modules are powered via the motherboard.If emergency control (on motherboard A) is activated, operation using the appropriate nominal video and audio powers is maintained even when the CCU is defective or not available.

Setting the "Video" Reference Voltage: Turn the potentiometer R24 on the motherboard A until the LEDs H1 and H2 light up. Check by means of CCU reset.Setting the "Video" Reference Voltage in the Event of "VF Input Fail":Unplug cable with VF input signal from transmitter input. Turn the potentiometer R84 on the motherboard A until the LEDs H1 and H2 light up.Check by means of CCU reset. *Setting the "Audio" Reference Voltage:*Turn the potentiometer R23 on the

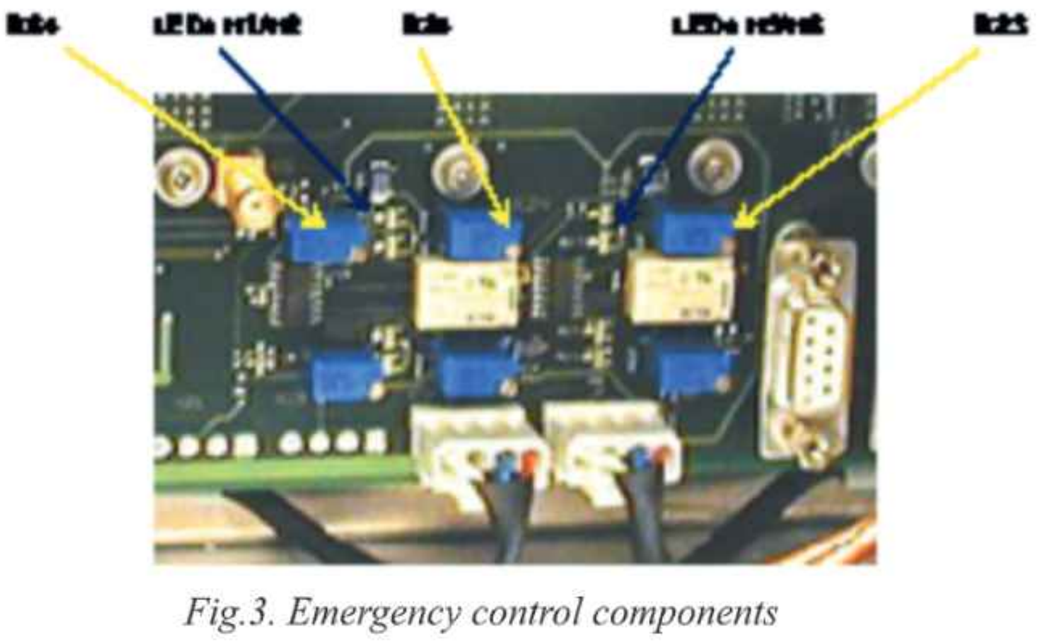


Fig.3. Emergency control components

motherboard A until the LEDs H5 and H6 light up. Check by means of CCU reset.

3.0. Starting "ON" sequenceof the TV Transmitter:

The basic mechanism for starting 'ON" of the transmitter and possible causes for failure to switch ON of the Transmitter is chronicled as below:

1. ON Command from TCU goes to Micro-Processer, and Micro-Processer instructs pre-decided Exciter A or Exciter B to switch ON. Before releasing the ON Commands, Micro-Processer checks whether the RF Loop is through or not.
2. If RF Loop is not through, then Exciter won't get switched ON. Then, check the RF Loop next.
3. If RF Loop is through, then ON Command is passed onto the Motherboard, where ON Signal is passed parallel to Solid State Relays K1 & K3, and K4 & K5.
4. Relay K4 & K5 gives the command to Modulator and Synthesizer through Rack Controller and Encoder to generate RF.
5. Relay K1 & K3 gives the command to Rack Controller. This signal then travels to Connector X51 in Power Panel Board.
6. Transmitter will fail to switch ON, if there are problems in:Cooling System, Power Supply system (Under-Voltage, Over-Voltage, Phase Sequence, Missing Phase), RF Loop.
7. In case upon giving the ON Command, transmitter does not get switched ON; and RF Loop is through, then the TCU is faulty.
8. Check Trans-Receiver IC D222 and D220 and replace the defective IC if any.
9. If TCU is not able to receive any commands.
10. Then replace both D222 and D220 IC's .

11. TCU shows broken Video Link: Either Encoder may be defective, Or TCU may have some problem.
12. If issue lies with TCU, then replace IC D22 (Embedded Controller). The glimpse of components of TCU is shown in Fig.4.

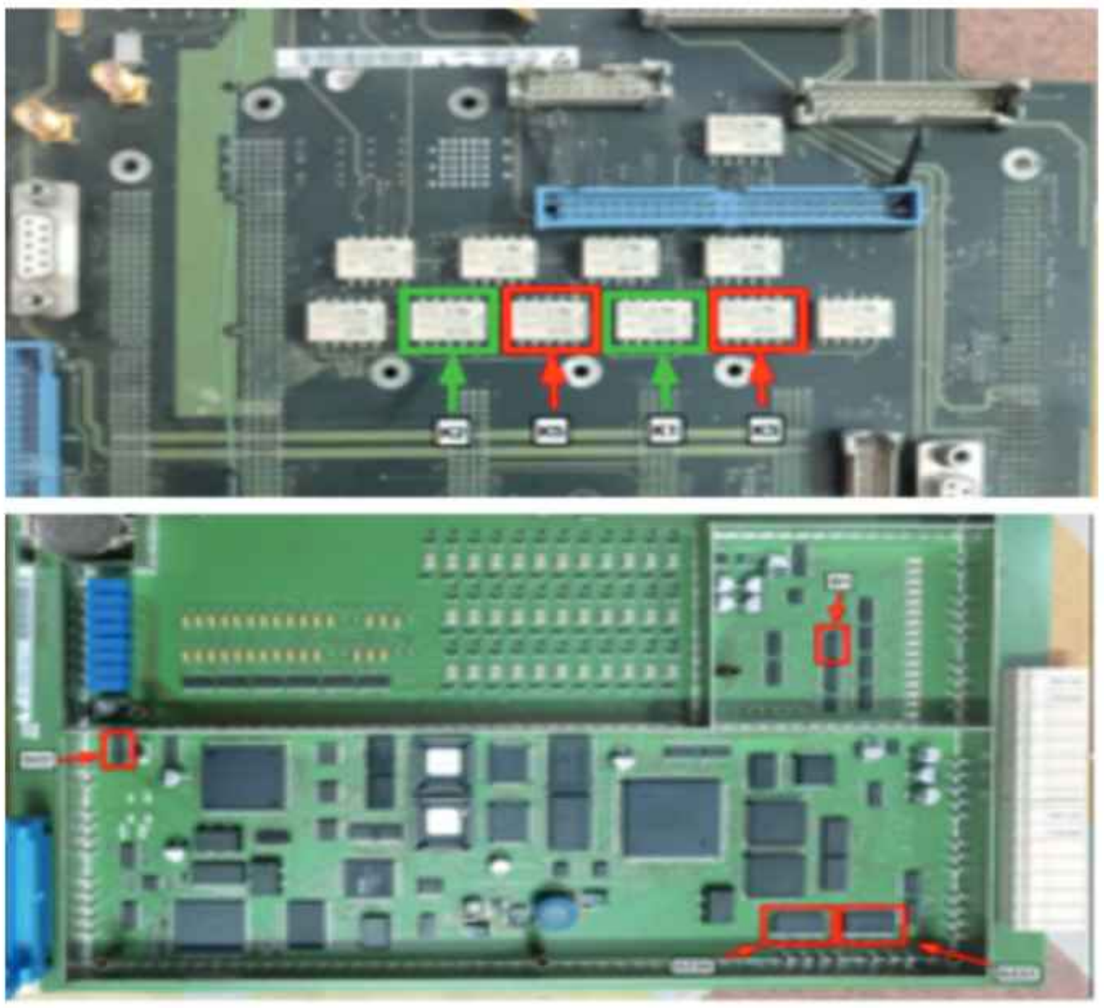


Fig. 4.Critical components of TCU

Fault	Component failure/Tests for Preamplifier
RF-IN Failure	Check Resistances R13(100Ω) , R17(100Ω), R17(681Ω), R407(100Ω), R460(4.75kΩ), R461(1Ω), R470(10 Ω),R444, R445 and R446(All 47.5 Ω) . Check IC U1, IC N-5,IC N-350 Check Voltage at Relay K1(11 V at pin 4,6,10. No volt at other pins) and Relay K2(12 V at pin 4,6,10 and 6 V at pin 8 & 9 . No volt at other pins.
RF Power Failure	Check Hybrid Amplifier ERA 5SM, Bipolar transistor V400 BGF 135, transistor V410 D1001, transistor V420 D1003, Check resistances: R534 and R535, (both 35.7 Ω Ohms),R536(301 Ω Ohms), R513(3.32 Ω), R514 33.2 Ω), R522(10 Ω), R521 & R523(511 Ω) , R524 (51.1 Ω), R525 & R526 both 1 Ω), R560 (10 Ω), R561 & R562 (both 10 Ω and R1(0 Ω)

Table 4. Component failures in Preamplifier

4. Power Amplifier: The VHF Power Amplifier R&S VM602A1 produces a sync peak power of 2.2 kW for analog TV or 485 W of DVB-T/T2. The broadband amplifier operates in the range 170 MHz to 250 MHz.The input signal is applied to the multi- stage preamplifier, which also comprises a power control circuit, amplitude and phase controls and several monitoring functions. It is followed by a driver amplifier and four output stage modules. The combiner and transformer networks are entirely in microstrip and

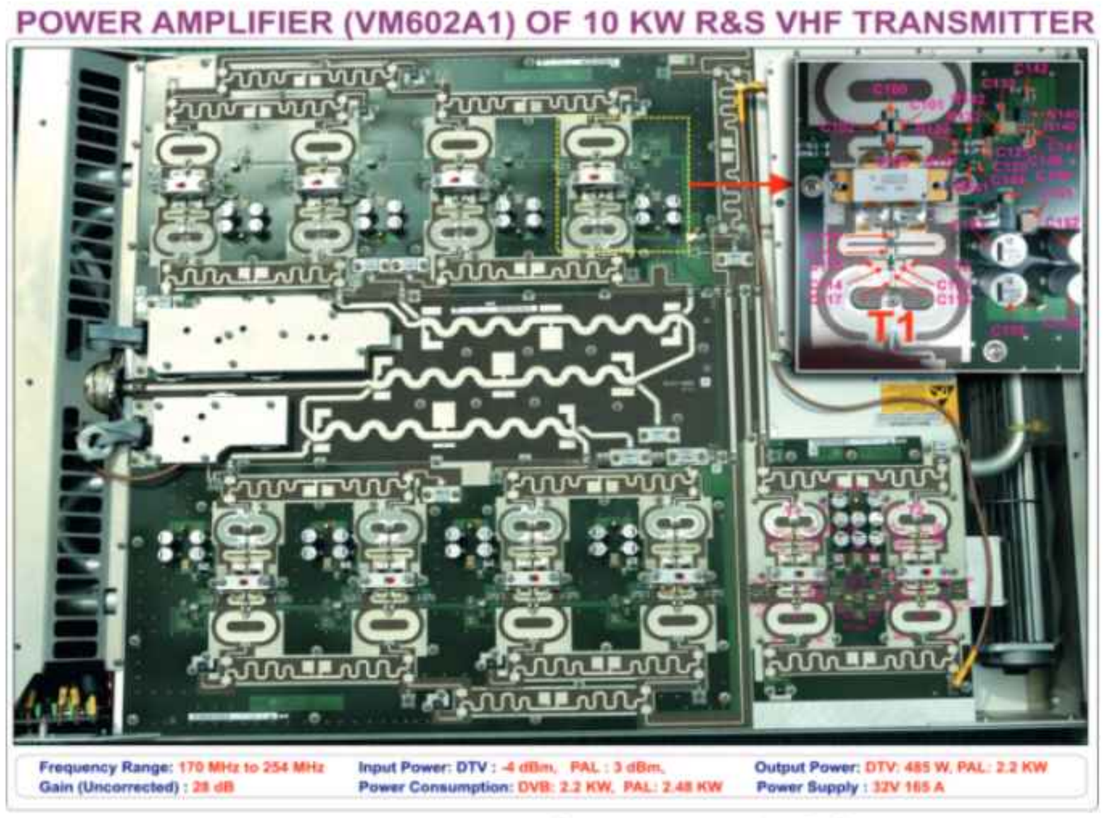


Figure 5. Anatomy of RF Power Amplifier

suspended substrate technology. Total gainis approx. 63 dB. With advanced LDMOS technology, the power amplifiers are characterized by high linearity, high efficiency and compact design. In case of a transistor failure, a special protective circuit prevents the other transistors from being over driven.

The operating point of the other transistors is not changed. The critical component failures in VHF PA are MOSFET 1007 and D 10030 and in case of UHF PA MOSFET 1007 and D 10010. The critical failures in Preamplifier are detailed in Table 4. The amplifiers can be used unmodified for digital and analogue TV. Like in the UHF version, there is a three-phase switching power supply fitted on the underside of the heatsinkin the amplifier module. An over temperature and reflection-responsive circuit plus an input level monitor are provided for amplifier protection. The transmitted output of the Transmitter depends upon total no of amplifiers/transistors and no. of faulty amplifiers/Transistors as shown in the formula.

$$P_{Antenna} = P_{Sum}\left(\frac{m-n}{m}\right)^2$$

Where $P_{Antenna}$= transmitted power, P_{sum}=max. available power and m=number of modules/ amplifiers/ transistors, n=number of defective modules/ amplifiers/ transistors.

The details of component failures in Power Supply and procedure of replacement of transistor is described as above. One axial cooling fan in Power amplifier also fails sometimes.

The other major failures in the transmitter includes the failure of RF Combiner and Reject loads due to Power mismatch, presence of spurious signals or lightening

etc. The RF resistors and strip line gets burnt in RF Combiners. The alignments of RF Powers and their displays and setting of critical parameters are described in the user manual of the transmitter and may be followed accordingly.

Replacement of a Transistor in the amplifiers: -

1. Switch off the amplifier and remove from the rack.
2. Remove the covers from the top and rear of the amplifier.
3. Terminate the amplifier in 50 Ohm, switch it on and without RF.
4. Connect AD Box 16 to the rear connector labeled Test port current and measure quiescent currents (0.5 V = 1 A).
5. The normal value for the driver transistor = 4A =2V. If the reading is 0 or 1000mV, a transistor may be defective. For the output amplifier transistor pair the value will be = 1.2A=0.6V. If the reading is 0 or 300 mV, a transistor may be defective.
6. Measure with ohmmeter (w.r.t ground) directly at gates(2 Transistors , 4 gates) of the presumed defective pair. If the resistance is clearly below 6 KΩ , either the transistor or the potentiometer is defective.
7. Remove the 2 Bias Resistors (100Ω) and measure the resistance to the ground.If the resistance is very high (of the order of MΩ), after removal of the Resistors, then the transistor is not defective.

Components failures in Power Supply of RF Amplifiers

i. MOSFET 18N120BND CY3
ii. 100R K9N
iii. TC 4424 CPA 35V
iv. Rectifier 143B341
v. Relay RT314012
vi. SMD IC UC3879
vii. Opto coupler CNY-2
viii. SMD IC LM2902D
ix. SMD IC LM 358
x. Rectifier Diode ISOTOP

5. Conclusions.

Repairs from OEM is costly and time consuming process and therefore It is crucial to gain capacity building for proactive actions to preempt the failure and optimal performance of the transmitters. The knowledge gained in workshop may help the Engineers to Condition monitoring key equipment's health indicators, equipment health checks, firmware upgradation, Preventative maintenance, Spares forecasting and forward planning to help manage equipment and spares availability etc. The critical components required for repairs of various modules have been listed in this paper. The HPTs may keep stock of such components for timely repairs. Hand on Trainings during the workshop facilitates confidence building and understanding the techniques and processes. Such augmented learning process are highly beneficial for the network.

6. Acknowledgement:

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iv. Sh. Mahak Singh AE, O/o ADG(NZ).
v. Inputs from Dharmendra Kumar SEA DDK Gorakhpur.
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About The Author



Mr. M. S. Duhan is aretired DDG(E) from Doordarshan. He has a rich experience of broadcastingtechnologies encompassing Studios, Earth stations andTransmitters of Radio and Television. He has been the keen student of RFWaves and has widely travelled in quest for study of RFwaves - AM, FM and COFDM. He has given manypresentations on Digital Terrestrial TV, Spectrum planning and FM Transmitters in various conferencesand seminars,as well as participated in TV Talk shows.

He has contributed significantly in the system design and implementation of DVB T2 TV Transmitters in India and 7th Plan of FM Transmitters in AIR. He isthe recipient of Doordarshan Annual Awards 'four times', for the best maintained High Power Transmitters& Maintenance Centre. With nearly four decades of experience in all broadcasting spectrum, he arranged various seminars. He was the resource person for AIBD/RTM in-countryworkshop on DVB-T2 at Kuala Lumpur.His technical articles were adjusted as "Best Article" and awarded by ABU during the years 2015, 2017, 2019 and 2020.

He orchestrated programme contents in Doordarshan channel "DD Bharti" for about 9 months, which is a dedicated channel to art and culture of Doordarshan.

Email: msduhan.de@gmail.com . Mobile: +919013462293.

Significance of Frequency band for success of Digital Radio Broadcasting in India

P. Das

Several Digital Radio Broadcasting systems have been developed for different frequency bands. While these systems share many common features, they differ slightly in aspects such as channel capacity, data service volume, and related parameters. In principle, any digital radio system can efficiently deliver radio services along with value-added features. However, the success of digital radio depends on the careful selection of the most appropriate system, based on national requirements and broadcasting priorities.

A primary prerequisite for introducing digital radio is the availability of spectrum for a nationwide coverage plan. Ensuring nationwide coverage, together with Single Frequency Network (SFN) planning, is necessary for successful deployment. Therefore, the choice of system must be guided by spectrum availability within the desired frequency band, as well as the country's needs regarding channel capacity and supplementary services.

Although all systems support SFN, which maximizes spectrum efficiency, this advantage may not be fully realized in frequency bands already heavily allocated to analogue radio. In such cases, digital radio may not yield optimal spectrum dividends. It is, therefore, essential to explore frequency bands where a nationwide digital radio network, including SFN, is feasible, and to identify systems most suitable for deployment in compliance with ITU Radio Regulations.

India, with a population exceeding 140 crore, is uniquely diverse in terms of languages, dialects, and cultural identities. Many states have their own languages and dialects, while several cities are cosmopolitan and multilingual. To serve this diversity, broadcasters must produce content in multiple languages and formats. This explains why India broadcasts a far greater number of radio and television channels compared to many other countries. Moreover, listeners consistently demand wide variety of programme. To meet these expectations, broadcasters will need to transmit at least 15 to 20 different channels per city in India. This requirement underscores the need for frequency bands capable of accommodating a large number of channels. Hence, the rollout of digital radio in India should prioritize bands that can support maximum channel capacity, ensuring both diversity of content and nationwide accessibility.

Any digital broadcast standard can be successfully tested as a pilot project. However, this does not guarantee its nation-wide rollout if spectrum availability and coverage planning are not done at the initial stage. Therefore, the availability of spectrum in broadcasting bands (MW, FM, and VHF Band-III) must be examined before deciding on any standard.

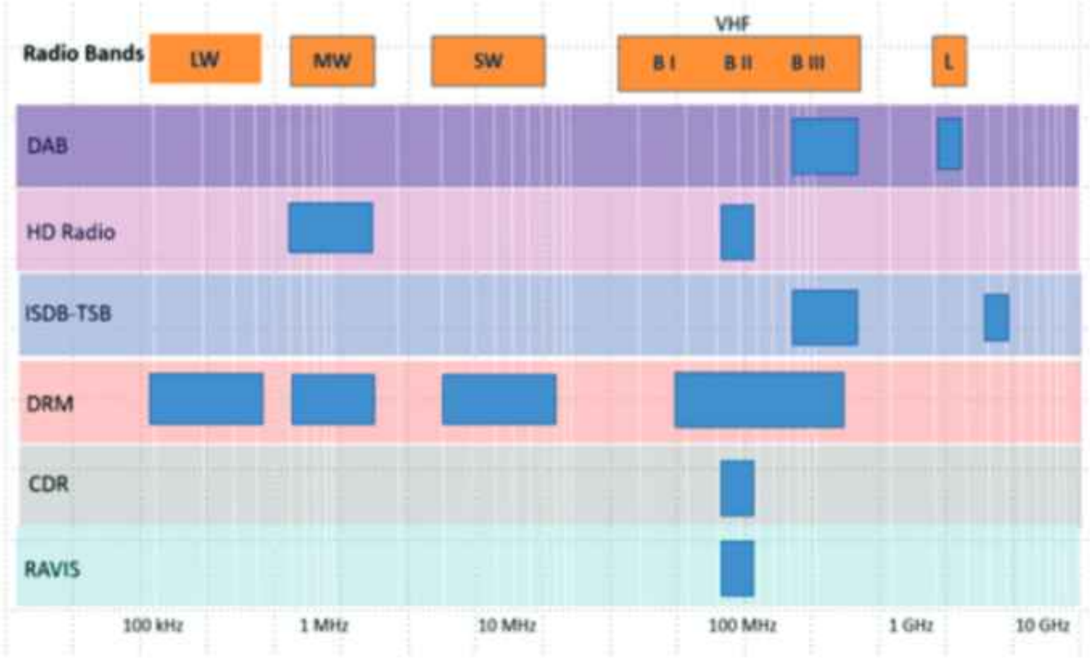
ITU recommended Digital Radio standards :

A number of digital radio broadcasting standards have been developed around the world. However, following digital terrestrial radio broadcasting standards have been recognized by the International Telecommunication Union (ITU) for various radio frequency bands:

ITU recommended Digital Radio Systems	Digital Radio Standards	Frequency Band
Digital System A	DAB/DAB+	VHF Band-III L Band (1452-1492 MHZ)
Digital System C	HD Radio	VHF Band-II/FM (88-108 MHz) & MW
Digital System F	ISDB-TSB	VHF Band III (174 – 230 MHz) & 2.6 Ghz
Digital System G	DRM/DRM+	VHF Band-II/FM (88-108 MHZ) VHF Band-III (174-230 MHz) MW & SW
Digital System H	CDR	VHF Band-II/FM (88-108 MHZ)
Digital System I	RAVIS	VHF Band-II/FM (88-108 MHz)

Frequency bands for Digital Radio broadcasting :

In terms of radio coverage in various frequency bands, Short Wave band (SW) provides the maximum coverage, followed by medium Wave (MW), VHF(FM) and VHF(Band-III). Therefore, to provide nation-wide



coverage, broadcasting in SW &MW bands would require lesser transmitters and CAPEX than VHF(FM)and VHF (Band-III). However, due to limited availability of spectrum these bands (MW & SW) can accommodate less no. of channels.

Availability of spectrum in various frequency bands for Digital Radio Broadcasting

1. Medium Wave (MW) (525 to 1605 Khz):

Frequency planning for radio broadcasting in the LF/MF bands in Regions I and III was developed by the ITU under the LF/MF Plan (Geneva, 1975). The MF plan covers the frequencies from 531 to 1602 kHz with 9 KHz channel spacing, assigning MW frequencies to various cities along with transmitter power levels for nationwide coverage. The plan entered into force in 1978 and remains valid today, with minor modifications achieved through mutual coordination among countries. Compliance with the plan has been widespread and enduring.

Within the Medium Wave band, two digital radio standards have been developed:

- Digital System C (IBOC – In Band On Channel)/ HD Radio
- Digital System G (DRM – Digital Radio Mondiale)

According to the ITU MF plan, the number of high-power MW frequencies (typically 50 kW or more) allocated per city varies across countries. In India, the availability of such high-power MW transmitters is limited, and all assigned frequencies have already been in use for analogue broadcasting.

When All India Radio (AIR)/ Prasar Bharati decided to introduce digital radio in the MW band using the DRM system, it adopted a two-phase rollout strategy:

(i) Replacement of ageing MW transmitters that had exceeded their operational life with DRM transmitters.

(ii) Upgradation of remaining MW transmitters still within their operational life to DRM capability.

Under this plan, AIR replaced about 37 high-power transmitters ranging from 20 kW to 1000 kW, achieving coverage of approximately 60% of the country. However, due to the non-availability of spare frequencies at these locations, DRM transmitters had to operate on the existing MW frequencies.

To retain analogue listeners in the absence of DRM receiver penetration, AIR adopted simulcast mode, broadcasting both analogue and digital radio on the same frequency, keeping digital power 14 to 17 dB below analogue power. This approach significantly reduced digital coverage and limited only two digital channels per transmitter, far fewer than what is required to attract listeners to invest in digital receivers.

Consequently, the limited digital coverage, restricted channel capacity, and the high cost of digital receivers have been key factors behind the poor success of digital radio in the MW band in India.

Countries who have spared frequencies inMWband may deploy Digital radio broadcasting avoiding simulcast. Advantage of this MW band is that it provides larger coverage with lesser CAPEX. However, use of SFN for regional or state-wise coverage may not be feasible due to spectrum constraint.

Issues with Simulcast:

- Simulcast is a bit complex for tuning / matching of RF sections and Antenna, particularly, in MW band
- Digital signal interferes with the analogue transmission leading to degraded audio quality in receivers. To avoid such interreference the Digital carrier is to be kept around 14 to 17 dB below the analogue power. However, it reduces digital radio coverage significantly.
- Analogue coverage also shrinks. This shrinkage affects listeners who rely on traditional analogue broadcasting
- Provides less no. of services/ channels thereby reducing choice before listeners.
- Simulcast is not considered an attractive proposition due to the technical challenges, interference issues, and limited service options.

2. VHF Band II/ FM (88-108 Mhz)

Each country has developed its own spectrum plan for FM transmission in the VHF Band II asper the planning parameters defined in Radio Regulations by ITU-R. Some of the parameters are not uniform across the world. For example, 100 kHz channel spacing assigned to Europe and many Asian countries including India and AIR has prepared its FM plan accordingly, whereas United States uses 200 kHz channel spacing.In Japan, FM broadcasting begins from 76 MHz onwards whereas in other countries it starts at 87 or 88 MHz.

In India, the FM band 88–108 MHz is used for multiple broadcasting services, which includes AIR/Prasar Bharati, Private FM broadcast, and Community Radio Services (CRS). AIR/Prasar Bharati primarily operates between 100 to 104 MHz, with a few exceptions in some metro cities. The remaining frequencies are allocated to private broadcasters and CRS. As per FM broadcast policy, Private broadcasters obtain license through auctionprocess and successful bidders are granted 15-year license for FM radio service in India.

Several digital radio standards have been developed for broadcasting in the FM band. These are :

- Digital System C (IBOC/HD Radio) – uses 200 kHz channel spacing
- Digital System G (DRM – Digital Radio Mondiale) – operates under 100 kHz spacing
- Digital System H (CDR – Convergent Digital Radio)
- Digital System I (RAVIS – Real-time Audio Visual Information System)

Identifying spectrum for digital broadcasting in the FM band is complex, particularly, while analogue FM expansion continues. Some of the key challenges in this bandinclude:

- Limited spectrum availability: Due to existing analogue FM broadcasting, frequencies for digital radio are available discretely in this band, which makes a nationwide digital plan difficult.
- Single Frequency Network (SFN) constraints: Although SFN is a major advantage of digital radio for spectrum efficiency, planning such networks is not feasible without availability of single frequency across regions or states.
- Suboptimal utilization: Given these constraints, the FM band (VHF-II) may not provide optimal spectrum efficiency for digital radio rollout in India.

Countries with lower demand for digital radio channels may find the FM band suitable for digital radio deployment. However, in India, where demand for diverse content and multiple channels per city is highdue to language diversity, the FM band is unlikely to meet long-term digital broadcasting requirements.

3 VHF Band-III (174-230 Mhz)

Historically, the VHF Band-III (174–230 MHz) was extensively used for terrestrial television broadcasting, and countries have developed detailed frequency plans for this purpose. In India, Doordarshanuses this band for analogue terrestrial TV services. However, following the closure of analogue terrestrial television transmissions, Doordarshan has no further expansion plan in the VHF Band-III.

Globally, the following digital radio broadcasting standards have been developed for operation in the VHF Band-III:

- Digital System A – Eureka-147 Digital Audio Broadcasting (DAB / DAB+)
- Digital System G – Digital Radio Mondiale (DRM), including the DRM+ mode

In addition to its current utilisation by Doordarshan in the VHF Band-III, there is considerable scope for accommodating digital radio services. A nationwide spectrum plan for digital radio broadcasting can be easily formulated within this band. VHF Band-III is particularly well suited for the deployment of Single Frequency Networks (SFNs) on a regional or cluster basis, enabling optimal spectrum utilisation, improved coverage uniformity, and efficient frequency reuse.

The Telecom Regulatory Authority of India (TRAI), in its recommendations on "Issues related to Digital Radio Broadcasting in India", under paragraph 3.11, observed that:

"Due to the non-availability of additional frequencies required for the expansion of analogue FM radio broadcasting services in the VHF-II band, most of the European countries are operating digital radio broadcasting services in the VHF-III band."

In Europe, Channel-12 of VHF Band-III has been specifically designated for digital radio broadcasting. Similarly, in India, vacant TV channels in VHF band III can be identified for digital radio through coordination with Doordarshan and WPC wing.

VHF Band-III emerges as the most suitable band for digital radio broadcasting in India. The availability of contiguous spectrum makes it feasible to plan nationwide digital radio networks, including large-scale SFNs, with minimal spectrum requirement.

For the Public Service Broadcaster, this band offers significant advantages. A single 7/8 MHz television channel slot in Band-III can accommodate at least four DAB+ multiplexes, or several DRM+ blocks, supporting approximately 70–80 radio services. Furthermore, since most of the existing television transmission towers already have vacant Band-III infrastructure, the installation of Band-III antennas can be undertaken with minimal additional capital expenditure.

All India Radio/ Prasar Bharati could utilise this capacity to broadcastits own services in a city including some private radio channels to add more variety to content.Private radio channels can be includedthrough a transparent auction process,similar to DD FreeDish,thereby generating substantial revenue while promoting diversity and competition in digital radio services. Under such scenario private broadcasters may not have to invest on CAPEX in terms of transmitter, antenna, etc.

Conclusion:

The demand for radio channels in a city is strongly influenced by the linguistic diversity of its population. Cities with a larger number of spoken languages require additional channels to effectively cater to varied linguistic communities. In addition, the availability of a wide range of programme formats addressing the interests of different age groupssuch as children, youth, working professionals, and senior citizensfurther increases the requirement for multiple radio channels/ services in a city.

A digital radio system that can accommodate desired no. of channels to provide multilingual broadcasting, diversified programme content, and value-added services (such as text information, emergency alerts, and data services) is more likely to succeed in the Indian context. In practical terms, a digital radio platform capable of delivering 15–20 radio channels per city would provide sufficient content choice to attract listeners and encourage the adoption of digital radio receivers.

However, achieving such service capacity requires adequate spectrum. A careful assessment of spectrum availability is therefore essential to accommodate the desired number of radio channels and services while ensuring uninterrupted and high-quality broadcasting. The frequency band selected for digital radio rollout should be capable of supporting the required channel capacity including optimal utilisation of spectrum. Correspondingly, digital radio technologies optimised for that band should be adopted to minimise capital expenditure (CAPEX) and overall network costs.

Further, comprehensive nationwide network planning, including the deployment of Single Frequency Networks (SFNs) wherever feasible, can significantly improve spectrum efficiency and coverage uniformity.

Simulcast operation of analogue and digital radio services should be minimised or avoided, as it limits digital coverage andaffects the quality of digital radio services. A well-planned, spectrum-efficient, and content-rich digital radio ecosystem is therefore critical to ensuring the long-term success of digital radio broadcasting in India.

About The Author



Mr. Padarabindadas, with over 33 years of distinguished service in Prasar Bharati, Sh. Das has held diverse roles across the broadcasting and media spectrum. His expertise covers installation of Radio and Television projects, broadcast spectrum planning and coordination, broadcast infrastructure planning, training, international relations, commercials, corporate affairs, etc. From 2007 to 2011, he had the privilege of working with the Asia-Pacific Institute for Broadcasting Development (AIBD), Kuala Lumpur, which enriched his perspective on international advancements in broadcasting technologies and innovations.

An Overview of the Third Generation AVS Video Coding Standard (AVS3)

Zhou Yun

Abstract:

This paper presents a comprehensive introduction of AVS3, which was developed by theAudio and Video Coding Standard (AVS) Workgroup of China and is published as GY/T 368by NRTA in 2023. To improve the coding efficiency, AVS3 adopted many novel technologies including block partition, intra/inter prediction, transform, quantization and entropy coding and has 50% bitrate savings compared to its predecessor AVS2.AVS3 has been used in China's 8K UHDTV channels and will have more worldwide applications in the future.

Introduction

In recent years, China has undergone rapid development in the UHD video industry. With higher resolutions, framerate, bit depth, dynamic range and wider color gamut, the amount of data of UHD video has increased largely. To meet the demand of 4K UHD video services, NRTA published the second generation AVS video coding standard (AVS2□in 2016. AVS2 has a similar coding efficiency with H.265 and is adopted for 4K UHD video coding in China with the bitrate of 36Mbps under the configuration of 3840×2160@50p, 10bit and HDR. However, 8K UHD video brings more challenges and it is necessary to develop new video coding standards tofurther improve coding efficiency.

AVS3's standardization process started in December2017. The aim of AVS3 is to have 50% bitrate savings with AVS2 under the premise of the same video quality. To promote industrial application, the development of AVS3 is divided into two phases: In the first phase, the primary consideration for the selection of coding tools is the limited decoder complexity, and it was finalised in July 2019 after two years of efforts.

Meanwhile, the second phase aims to improve coding efficiency based on the first stage. AVS3 wasfinalised in 2022. After a comprehensive evaluation and testing procedure,AVS3 was adopted as the broadcasting industry standard in China and was published by NRTA in March this year. This paper gives an overview of the AVS3 standard including key technologies and its application in China.

Key coding tools of AVS3

AVS3 adopts a block-based hybrid video coding framework as shown in Fig.1. A flexible block partition mechanism is put forward to adapt to higher resolution and novel intra/inter prediction, transformation, quantization -- entropy coding tools are also introduced to highly improve coding performance.

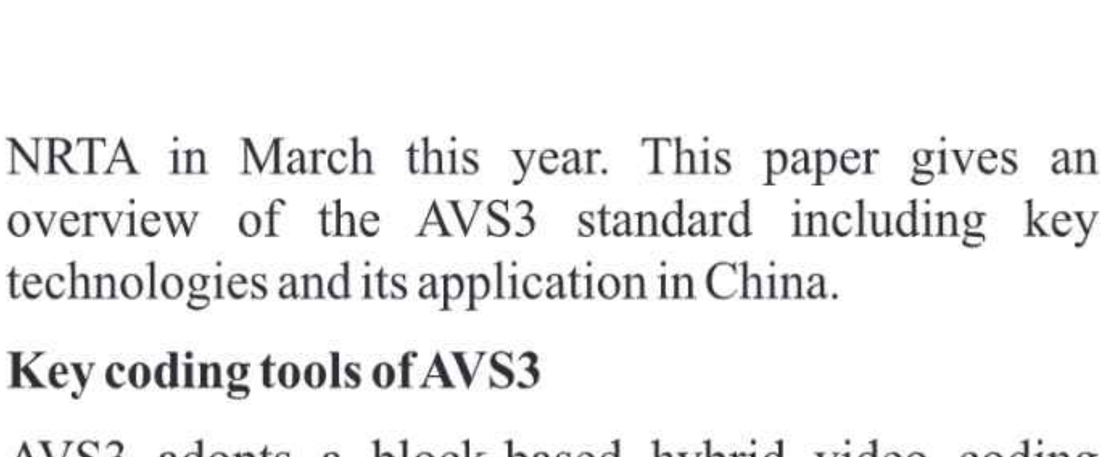


Figure 1 Coding framework of AVS3

1. Flexible block partition structure

AVS3 designs a more flexible block partition structure to adapt for different video resolutions. The coding picture is firstly divided into patches and then are divided into largest coding unit (LCU). The

largest size for LCU is 128x128 which is more applicable for UHD videos. LCUcan be recursively split into coding units (CU□based on partition tree structure. AVS3 adopts quad-tree (QT), binary tree (BT) and extended quad-tree (EQT) partition to decide the final size of CU as shown in Fig.2.

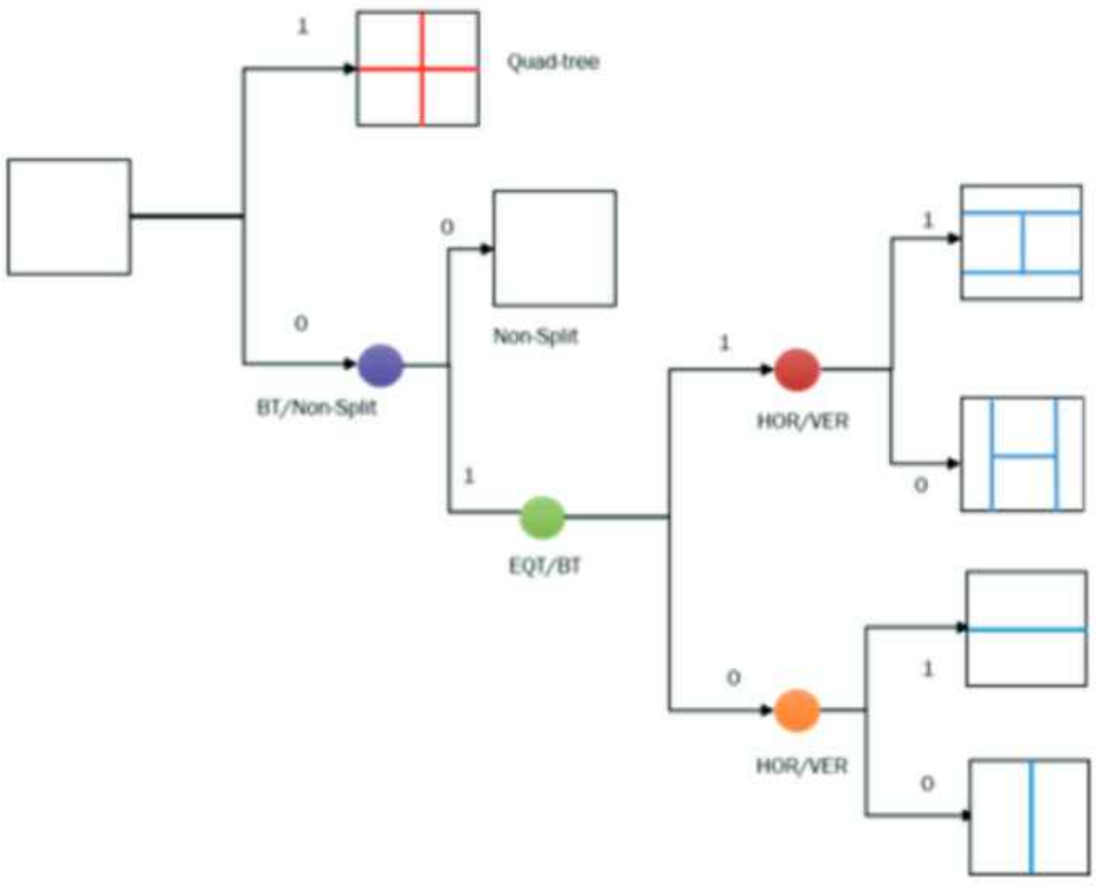


Figure: 2 illustration of block partition in AVS3

For each LCU, one bit is first used to indicatewhether the LCU is divided into QT or not. If not, one morebit is then used to represent whether the LCU is divided intoEQT or BT. Through this flexible partition process, CU can be non-square.

2. Intra prediction

AVS3 adopts 66 luma intra prediction including 62angles prediction modes, DC mode, planar mode, PCM mode and bilinear mode. AVS3 introduces Intra DerivedTree (Intra-DT), intra prediction filter (IPF) and two-step cross-component prediction mode (TSCPM) to improve intra-prediction.

Intra Derived Treedefines intra coding unitpartitioning mechanism vertically or horizontally. Thecurrent CU is divided into two or four prediction units (PU)when the current CU apply to intra-DT. Intra-DT has 6 split types.

IPFis a coding tool to improve the accuracy of the intra predicted pixels. The predictedvalue near the top and left boundary of current CU is filteredto enhance the prediction accuracy.

TSCPMis designed for intra chroma coding, which assumes thelinear correlation between the prediction results of luma andchroma components. The linear regressionfunction is first applied to the originally reconstructed lumablock to get an internal prediction block. Secondly, the internalprediction block is sampled down to generate the final chromaprediction block.

3. Inter prediction

In AVS3, the Affine prediction mode considers more complexmotion instead of translational motion. There are two types ofaffine motion predictions, one is affine direct mode whilethe other is affine inter mode. In affine direct mode, an affine candidate list, including atmost 5 affine models, is derived from the motion information ofneighboring blocks,an affine model candidate is a 4-parameteraffine model or 6-parameter affine model, represented by 2 or3 motion vectors.When the CU is affine inter mode, it is 4-parameter affine motion model.

Adaptive Motion Vector Resolution (AMVR) is adopted to reduce the coding bits for the motion vectordifference (MVD) coding. The MVD is usually coded with1/4-pel precision. AMVR allows various resolutionsfor MVD coding. For instance, 1/4–pel, 1/2–pel, 1–pel, 2–peland 4–pel resolution are supported.

Ultimate motion vector expression (UMVE) technologyis an effective way to express motion vector for skip and directmodes in AVS3. According to the base motion vectors derivedfrom the information of spatial or temporal neighboring candidates, this technique offsets these motion vectors to obtain themotion vectors that are more suitable for the current codingblocks.

Based on the above HMVP and AMVR, an enhanced motion vector resolution (EMVR) mode is carefully designedto avoid the shortcoming of one motion vector prediction(MVP) compared to multiple MVP. EMVR is proposed tobind the AMVR and HMVP, which corresponds to the existingAMVR index of current CU with the motion information inHMVP table, and then conducted inter-frame prediction.

Symmetry motion vector difference (SMVD) is usedfor motion vector difference in inter Bi-directional predictionmode. One bit is used to indicate whether SMVD is adoptedwhen the prediction mode of current CU is Bi-directionalprediction mode. When SMVD is used, the motion informationin List1 is symmetrical with respect to List0.

4. Transform

Position-based transform (PBT) is introduced in AVS3, DST-VII and DCT-VIII are added to the transform module. The selection between DCT-II, DCT-VIII and DST-VII depend on signaling bit and intra block's position. AVS3 introduces a newtransform method which uses pre-designed transform sets forfour sub-blocks according to their positions. A new CU-levelsyntax element, PBT flag, is introduced. When PBT flag is0, the traditional DCT-II transform is used for current CU.When PBT flag is 1, PBT splits the CU into four sub-blocks,and uses pre-designed transform set according to their position. In each sub-block, itsposition number relates to a pre-designed transformset.

5. In-loop filtering

Enhanced sample adaptive offset (ESAO) is applied to consider the textural and edge directional features. There are two feature descriptors in the sampleclassification process. The first one is calculated based on local binary patterns by comparing the intensities of the current sample with its 8 neighboring samples. The second descriptor is derived based onthe amplitudes of the current reconstructed samples .

Cross-component sample adaptive offset (CCSAO)is only applied for chroma components.There are two feature descriptors in the CCSAO classificationprocess. The first one is based on the difference between the corresponding luma sample and its neighboring samples. The seconddescriptor is based on the amplitude of the current sample or its corresponding luma sample.

Besides, AVS3 improves adaptive loop filter (ALF), a larger filter shape is enabled with 15coefficients and 29 taps. To make the filter more adaptive, the reconstructedframe is divided into 16 or 64 categories.

Coding performance

In order to evaluate the coding performance of AVS3, a subjective assessment between AVS3, AVS2 and Versatile Video Coding (VVC) wasconductedfor 8K testing sequences with the parameters of 7680×4320@50P, 10bit, BT.2020, HLG. And the coding bitrate is set 72Mbps, 144Mbps and 72Mbps for these three standards.

At the given encoding rate, AVS3 has 5.3% quality reduction compared with the original sequences, while AVS2 and VVC have5.0% and 5.3% quality reduction separately. The testing results shows that AVS3 hassimilarcoding performance with VVC and can reach 50% bitrate savings with AVS2.

8K Test sequences	Quality Reduction		
	AVS3@72Mbps	AVS2@144Mbps	VVC@72Mbps
1	4.6%	3.6%	4.0
2	6.7%	6.7%	7.7
3	4.3%	3.5%	4.6
4	5.7%	5.6%	4.1
5	7.5%	6.2%	7.2
6	3.4%	5.0%	5.6
7	6.3%	5.5%	4.4
8	3.8%	4.0%	5.0
Average	5.3%	5.0%	5.3%

TABLE I Subjective Assessmen to fAVS3

Industry Applications

The AVS3 has attractedtremendous attention from the industry ever since the finalisation of the AVS3 Main Profile in March, 2019.The designation and performance-complexity trade-off naturally fulfill the demands and requirements ofdifferent use cases. AVS3 has deployed from industrial-grade 8K real-time software codecs to 8K decoder chip for smart-phone, from 8KUHD TV broadcasting service to worldwide large-scale events support.

Based on the Intel scalable video technology (SVT) architecture,a novel AVS3 8K real-time software encoder has been presented, namely SVT-AVS3. The SVT-AVS3encoder achieves an optimal trade-off among coding performance, encoding speed, and perceptual visual quality.

The AVS3 all-platform open-source decoder "MERAK" has been released. The decoder was developed by the team from Peking University Shenzhen Graduate School, code-named uavs3d, and which supports up to 8K/60P real-time video decoding.

In September 2019, Hisilicon Inc. officially released a new end-to-end solution for 8K content based on the world's first AVS3 8K decoder chip Hi3796CV300 in International Broadcasting Convention (IBC), which greatly promotes the 8K applications of AVS3.

On February 1, 2021, the China Media Group launched an 8K UHD test channel using the AVS3 video standard. On January 1, 2022, the Beijing TV Winter Olympics Documentary Channel was broadcast using the AVS3 video standard. On January 25, 2022, the 8K UHD channel of China Media Group (CMG) conducted its broadcast using the AVS3 video standard.

In July 2022, the AVS3 officially became one of the next-generation video codec standards in the DVB standard system, which means that AVS3 video standard, as an option of video encoding of international organisations, will greatly support the future development of the global UHD video industry.

Conclusion

AVS3 is now opening a new era for 8K UHD video coding in whichend-to-end solutions and commercial-level AVS3 codecs have been incorporated and established. Severaldirections are currently under investigation for future research and applications, including neural-networkbased video coding technology and machine-vision oriented video coding.

** Courtesy ABU for providing the document to publish in BES Review*

About The Author



Ms. Zhou Yun, Senior Engineer, Academy of Broadcasting Science (ABS), National Radio and Television Administration (NRTA), China

Zhou Yun, received the B.S. and M.S. degree from Beijing University of Posts and Telecommunications. She is currently a senior engineer at Academy of Broadcasting Science, NRTA. Her research interests include video and audio coding, UHDTV and immersive media.

SVURG/Z-Mod – An Introduction

Satyapal & Rakesh Aggarwal

SVURG-Z-MOD technology presented here provide a new horizon, opening up enormous data capacity enhancement in existing spectrum by adopting cycle by cycle modulation of the carrier frequency. It not only provides thousands of times more data capacities by removing bandwidth consuming "SIDE BANDS" and make carrier itself carry one byte of information within each carrier cycle. This technology providesopportunities to establish all new secure communication standards, by integrating real time dynamic handshake protocols and processes.

This paper describes a new, patented, technology named **SVURG-Z-MOD**, which reduces spectrum needs common to all present-day forms of analogue and digital transmission, by removing side bands completely to carry **"many times"** more information by carrier itself. Out of many advantages of this Z-MOD technology, simultaneous accomplishment of three prominent changes (unbelievable) are worth noting: -

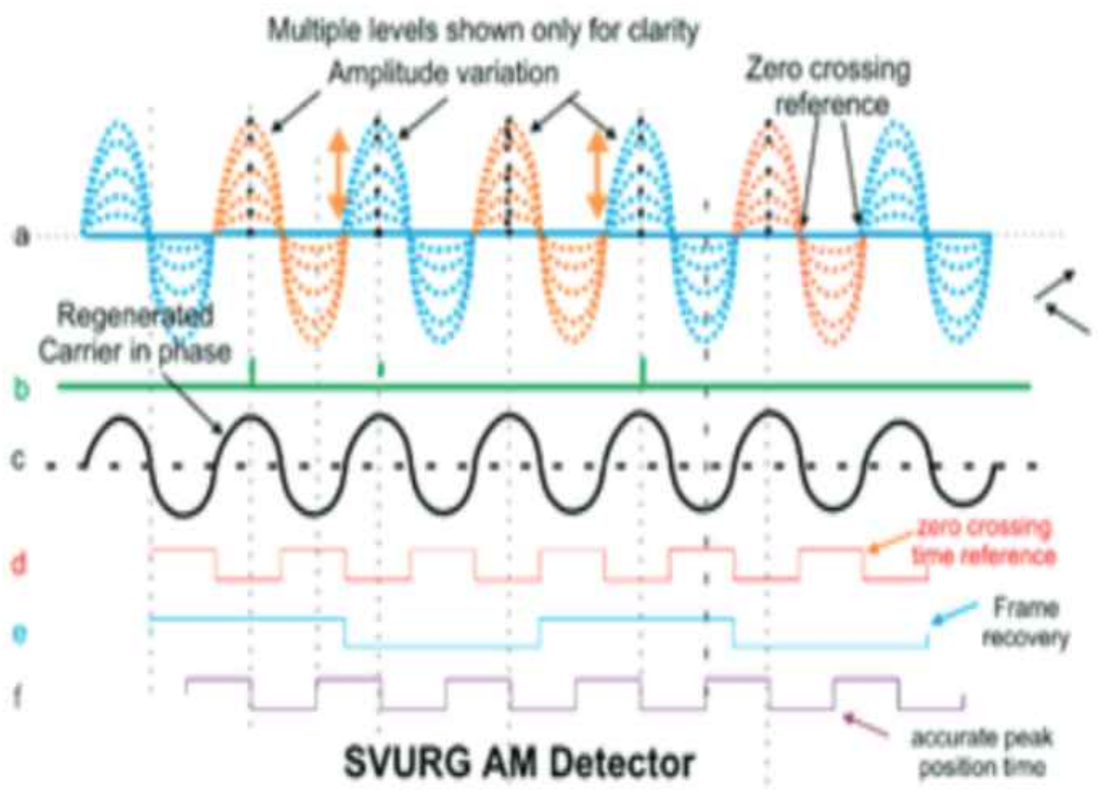
1. Conservation of spectrum usage to just the carrier frequency itself (achieving theoretical limit of Zero bandwidth).

2. Information carrying Capacity increased by many times.

3. Reduction in radiated power requirement to less than 0.1% as compared to conventional methods.

This technology provides a new method of modulation process called **"Z Mod"**, which improves spectrum efficiency by removing side bands completely, unleashing unlimited possibilities. The patented **SVURG** technology provides true inexhaustible bandwidth capacity to carry many times more signal/data, improve quality of communication with negligent noise and take all future communications to next generation serving mankind.

To establish that large amount of data can be carried by carrier pure sine wave cycles without need of side bands, consider Amplitude Modulation wave form. Let us assume carrier sine wave shown in above *Fig-a* are pure sine wave cycles and conventional diode detector will only pass the peak of each carrier cycle to filter to recover say 1 KHz audio signal from 455 KHz IF signal. Here the point to note is that, for filter, it will not make



any difference whether the IF signal frequency is '455 Khz' or '455 ± 1 KHz' to reproduce 1 KHz signal.

In SVURG systems the frequency becomes constant and there is no variation in frequency due to modulating signal, but amplitude of each cycle can differ from previous cycle. In synchronous detector as shown above we only take a sample at peak position of 455 KHz signal resulting in data values at 455 KHZ sample rate and resolution of each sample can be up to 24 bits with existing technology. This is equivalent to 10.92 megabits per second, which is phenomenal increase in capacity. In analogue terms Nyquist theorem says twice the highest signal frequency samples are needed to reproduce it. Practically we have seen analogue signal of ⅓ rd. of sample frequency can be reproduced very nicely. This calculates to 455/3=151 KHz of analogue signal bandwidth can be reproduced by the above SVURG method.

To solve our assumption of carrier being sine waves, **"Z Mod"** concept, is capable of generating sidebands free modulated carrier waves, It generates each individual carrier sine wave cycle 0° to 360° independently, rather than continuous carrier wave. By Interpolating modulating signal at carrier frequency by sampling it at Fc, sample becomes static numeric value for each 0° to

360° carrier cycle duration and can be denoted by 'Δ'. This numeric value remains constant during each individual carrier cycle period and can change only in the next cycle.

Key point of **"SVURG/Z-Mod"** process is to ensure there is no change in sine property during the generation of each individual modulated carrier sine wave cycle. Change in sine property (say amplitude), according to sampled interpolated modulating signal numeric value takes place only at the zero-voltage crossing point, where the energy in cycle is zero and any change at this point, with different amplitude value results in net zero,but changing invisible amplitude vector (multiplication factor) for the new cycle to be generated with different sine cycle amplitude. Sine function remains constant during the entire sine cycle from 0° to 360° and only the Constant 'C' gets multiplied by 'Δ'.

equation

$C(t) = C \sin\omega c$ *.....carrier sine function*

becomes

$C(t) = 'Δ'C \sin\omega c$ *.....modulated carrier*

For each individual sine cycle being generated keeping function "sinωc" constant, resulting in generation of pure sine wave cycle without side bands, in cycle-by-cycle steps.

"SVURG/Z-Mod" process treats and generates each carrier cycle individually in a **cycle-by-cycle** process rather than generating carrier frequency as continuous wave. Another difference in SVURG approach is to convert continuous modulating signals into synchronised stepped static samples, interpolated at carrier frequency, with one single static value, for the entire duration of each carrier cycle. Next step is to apply this static value at zero crossing point of carrier cycle generation process, to change desired sine property for the duration of entire cycle, ending at another zero-crossing point. As a result, the carrier cycle is generated with constant sine properties for each complete carrier sine wave cycle.

The next cycle can have different sine property, changed only at the ending zero crossing point of previous cycle and start zero crossing point of new cycle. For example, in case of amplitude modulation the amplitude of new carrier cycle will differ from previous cycle according to interpolated static amplitude difference in sample derived from the modulating signal. Similarly in FM the 'Δ' value, will change frequency in steps retaining amplitude constant and cycle being pure sine wave of stepped frequencies. In phase modulation starting phase will change according to 'Δ' value in term of phase angle.

This unique method of using the zero-crossing point to change sine property with interpolated static value, results in removal of side bands completely. But as the generating parameter of sine wave cycle changes from one cycle to next cycle, each cycle can carry huge information, and can be defined by the **Nyquist Theorem**, where carrier frequency becomes the sample rate. This reduces, the needed spectrum bandwidth to fraction of 1 Hz. In absence of bandwidth consuming side bands, it further leads to restricting system noise to reach theoretical limits, increasing the signal to noise ratio as per Johnson noise theorem. Extra ordinary signal to noise ratio achieved in the process provides unimaginable data capacity as specified by **Shannon's theorem**, which sates, data capacity,"D" for specified bandwidth 'B' is directly related to signal to noise ratio in linear ratio: -

$$D = B \log_2\left(1+\frac{S}{N}\right) \text{ .....(I)}$$

Where S/N is a linear ratio of signal to noise.

It is further explained by well-established noise calculations, in terms of Minimum Detectable Signal MDS: -

*MDS (dBm) = 10Log (kTo*1e3) + NF + 10 Log (BW) + SNR (dBm) ...(ii)*

MDS (old) = -174 + 1.5 + 40 + 40 (dBm) = -92.5 dBm .....(iii)

Equation …(iii) is for a bandwidth of 10 KHz and a noise figure of Front end 1.5 dB with signal SNR of 40 db. resulting in -92.5 dBm absolute carrier level of MDS. For **"SVURG/Z-Mod"** S/N calculations bandwidth is 1 Hz and value of SNR in DB becomes 0, giving figure of 171.5dB

SNR = -174 + 1.5 + 1 + 0 (dBm) = -171.5 dB .....(iv)

Converting this result to linear number, it goes in millions (>140 millions) as per Shannon's theorem. This is a mind-boggling number using less than 1 Hz spectrum bandwidth. These are mathematical calculations and further validation by the scientific community is in process.

Conclusion

The **"SVURGZ-Mod"** concept claims of zero bandwidth, higher data capacity and fraction of a percent RF power requirement, appear unrealistic and unbelievably tall at first, in consideration of conventional teachings and beliefs. But as you corelate the concept in light of scientific explanations and calculations,based on two important principals established by Shannon's and Nyquist theorems; it unfolds like a miracle becoming true. Proof of concept prototype is already tested and available with the inventor. The concept is fully supported by theFourier transform in mathematical model,available on request in a detailed document.

About The Author



Mr. Satyapal With a career spanning over 35 years of dedicated service to All India Radio (AIR) and Doordarshan, Satyapal is a titan of the broadcasting industry. A Bachelor of Engineering (B.E.) by training, he has combined technical mastery with strategic leadership to shape the landscape of national media.Throughout his tenure, Satyapal has been the technical backbone of large-scale broadcast operations, managing the full lifecycle of radio and TV transmitters, studio environments, satellite earth stations, and complex RF systems.Core CompetenciesDigital Transformation: Leading large-scale transitions from analog to digital ecosystems.Strategic Maintenance: Implementing predictive strategies to ensure zero-fail transmission.Technical Operations: Expert management of Satellite Earth Stations and high-power RF systems.Project Leadership: Spearheading multi-regional infrastructure upgrades and optimizations.Key AchievementsMulti-Awarded Industry Expert: Recognized five times for technical brilliance at the highest levels:3x Winner of the Best Technical Paper Award at the Doordarshan National Awards.2x Winner of the prestigious Broadcast Engineering Society (BES) Award, honoring outstanding contributions to the field of broadcast technology.Digital Transformation Pioneer: Successfully spearheaded the modernization of analog transmission systems to digital platforms, significantly expanding network reliability and signal coverage across diverse regions.Operational Optimization: Developed and deployed advanced maintenance frameworks that drastically reduced system downtime and streamlined operational efficiency for national broadcast hubs.Infrastructure Excellence: Orchestrated high-impact infrastructure upgrades, ensuring the integrity of mission-critical broadcast services under demanding conditions.

Mr. Rakesh Aggarwal inventor of the SVURG/Z-Mod technology is a veteran Ex ISRO scientist and founder chairman of COMCON group of companies. He is known worldwide for his leading role in digitisation of radio network in India with supply and commissioning of 35 high power Nautel DRM transmitters for All India Radio. He has to his name more than a dozen national and international patents. His inventions include two disruptive technologies, one known as "SVURG/Z-Mod" for zero side band communication and other is the "Drag cum lift" wind turbine technology named "GURV", capable of generating more than 4-5 times more in green energy

News From Broadcasting World

Rajeev Kumar

China to host ITU World Radiocommunication Conference 2027 in Shanghai

The International Telecommunication Union (ITU) is pleased to announce that the next Radiocommunication Assembly (RA-27) and World Radiocommunication Conference (WRC-27) will take place in Shanghai, China, from 11 October to 12 November 2027.

Held every four years, the World Radio Communication Conference (WRC) reviews and revises the ITU Radio Regulations, the international treaty governing the use of the radio frequency spectrum and associated satellite orbits. During the four-week conference, ITU Member States will consider the results of technical studies and adopt decisions that shape the future use of spectrum for a wide range of radiocommunication services and

ITUWRC
WORLD RADIOCOMMUNICATION CONFERENCE

applications, including mobile, satellite, radiolocation, radio astronomy, and space research services that support lunar communications.

"WRC-27 will be a defining moment in making universal, meaningful connectivity a reality," said ITU Secretary-General Doreen Bogdan-Martin. "The conference will make critical decisions about sharing spectrum and satellite orbit resources efficiently and equitably, on Earth and in space, in ways that benefit all of humanity."

Over 4,000 delegates are expected to attend the WRC-27 and RA-27 meetings. In addition to ITU's 194 Member States, representatives from sister United Nations agencies, Regional Telecommunication Organizations, and ITU Radiocommunication Sector Members will participate as observers.

"World Radiocommunication Conferences are essential to shaping the future of global communications," said Mario Maniewicz, Director of the ITU Radiocommunication Bureau. "WRC-27 in Shanghai will guide the evolution of radiocommunication services on Earth and in space — for broadband connectivity, safety of life, and space and Earth observation. It is where the international community comes together to ensure that spectrum regulations keep pace with rapid technological innovation and respond to global, regional, and national needs."

WRC's inclusive and collaborative approach enables consensus-building among all stakeholders. It ensures that the Radio Regulations remain a stable, predictable, and universally applied framework – fostering an interference-free environment that supports continued investment in radiocommunication services.

(Source: https://www.itu.int)

Prasar Bharati- India launches 'Creator's Corner' to empower digital content creators

Prasar Bharati launched 'Creator's Corner' on DD News on 9 January 2026, providing a dedicated platform for digital creators across India. Information and Broadcasting Minister AshwiniVaishnaw described the initiative as the first step in major reforms planned for 2026, which will orient institutions such as Doordarshan and All India Radio towards new generation creators and technology-driven processes. He noted that the WAVES platform launched previously has already contributed nearly ₹5,000 crore to the creator economy whilst adding one crore youth and generating new employment opportunities.

Union Minister of State L. Muruganemphasised that the platform would empower creators from small towns who are independently producing content without large studios, aligning with the Prime Minister's vision for the Orange Economy. I&B Secretary Sanjay Jaju said the



initiative would build a vibrant and inclusive creator ecosystem, recognising creators as complete content makers rather than just performers.

The programme will be telecast at 7 PM from Monday to Friday on DD News, with repeats at 9.30 a.m. the following day. Each episode will feature four to six reels covering diverse themes including culture, travel, health, education, science, and entertainment. The initiative creates a mutually beneficial partnership, offering creators national reach whilst enabling the public broadcaster to curate innovative content that resonates with younger audiences.

(Source: https://mib.gov.in)

The 2nd Meeting of the APT Conference Preparatory Group for WRC-27 (APG27-2):

The 2nd Meeting of the APT Conference Preparatory Group for WRC-27 (APG27-2) was held in Pattaya, Thailand, from 28 July to 1 August 2025. It was organized by the Asia-Pacific Telecommunity (APT).



The meeting was attended by 704 delegates, of whom 380 participated in person. 30 APT Members, 1 Associate Member, 33 Affiliate Members, and 6 international organizations were represented at the meeting.

The objectives of the meeting were:

1) to finalise appointment of drafting group chairs for individual WRC-27 agenda item
2) to arrange briefing sessions on WRC-27 agenda items
3) to consider the progress in ITU-R Study Groups in relation to WRC-27 agenda items, and take necessary actions as appropriate
4) to develop APT preliminary views on WRC-27 agenda items based on input contributions from APT Members
5) to review the activities of other regional organisations, in particular, their preliminary views/positions with a view to fostering inter-regional cooperation
6) to review issues related to RA-27 and develop preliminary views of APT

As pending work of APG27-1, APG27-2 elected Mr. Christopher Worley (Australia) and Mr. Ali Reza Darvishi (Islamic Republic of Iran) as Vice-Chairs of APG-27. Mr. Zhao Zheng (People's Republic of China) was appointed as Special Adviser to APG-27, with the similar status of the Vice-Chairs. The meeting also appointed Mr. Bharat Bhatia (India) as Chair of the Editorial Committee of APG-27 with the similar status of the Vice-Chairs. APG27-2 also confirmed the appointment of Chairs/Co-Chairs of the Working Parties (WPs) of APG-27 which were temporarily decided at APG27-1. APG27-2 also approved the list of Drafting Group (DG) Chairs which were appointed by the various WPs of APG-27. The complete list of the Office Bearers of APG-27 and list of DG Chairs are available at APT website.

(Source: https://www.apt.int)

India Hosts the WAVES 2025; Over 90 Countries Participate in the Pioneering Initiative of Ministry of Information and Broadcasting

In 2025, the Ministry of Information & Broadcasting carried out many important initiatives to support India's media and entertainment sector. The biggest among them was WAVES 2025, the World Audio Visual and Entertainment Summit.

Prime Minister Modi described WAVES not just as an event, but as a "wave of culture, creativity and universal connectivity," and encouraged creators worldwide to



"dream big and tell your story." PM also emphasized India's vision of "Create in India, Create for the World," inviting global investors and youth to engage with India's expansive creative ecosystem.

Wave 2025 witnessed participation from over 90 countries, with more than 10,000 delegates, 1,000 creators, over 300 companies, over 350 startups, and an overall footfall of approximately 1 lakh attendees, spanning diverse sectors including broadcasting, infotainment, AVGC-XR, films, and digital media. WAVES platform is being continued through its three futuristic verticals:

1) CreatoSphere & Create in India Challenges (CIC): CreatoSphere is an immersive hub of innovation that places creators at the centre, transforming ideas into experiences across film, VFX, VR, animation, gaming, comics, music, broadcasting, and digital media. It brings together leading creative minds from India and abroad to foster dialogue, partnerships, innovation, and global showcasing of talent.

2) WaveX: WaveX aims to support and engage over 200 Startups as part of its initiative to strengthen the innovation and startup ecosystem. It enabled more than 30 startups to pitch to global industry leaders such as Microsoft, Amazon, and Lumikai, while nearly 100 startups showcased solutions through exhibition booths. A key milestone was the selection of VYGR News and VIVA Technologies (both supported by WaveX) for pitching on Shark Tank India, underscoring national recognition and credibility.

3)WAVES BAZAAR: WAVES Bazaar is the Global e-Marketplace for Films, Game Developers, Animation & VFX Services, XR, VR & AR Services, Radio & Podcast, Comics & E-Books, Web-Series, Music. Designed as a "craft-to-commerce" initiative, it connects Indian creators and institutions with global & domestic markets through curated festival/event participation, B2B meetings, co-productions, investments, and collaborations, in coordination with industry stakeholders. *(Source: https://mib.gov.in)*

The Meeting of the SATRC Working Group on Spectrum

The Meeting of the SATRC Working Group on Spectrum was held from 12 to 14 August 2025 in Maafushi Island, Maldives, hosted by the Communications Authority of Maldives. Mr. Ilyas Ahmed, Chief Executive of Communications Authority of Maldives, delivered the welcome remarks, followed by an inaugural address from H.E. Mohamed Kinaanath, Honorable Minister of State for Homeland Security and Technology, Maldives. The meeting brought together 41 delegates from SATRC Members and Affiliate Members, with 33 attending in person from 8 SATRC Member countries.



As part of the implementation of SATRC Action Plan Phase IX, the meeting considered the draft reports on 6 assigned work items. These reports were discussed in detail, and a number of modifications were proposed. The revised reports will be submitted to SATRC-26 as part of the outcomes of Action Plan Phase IX. The work items are:

- Spectrum approaches and regulatory requirements for NGSO satellite constellations for space-based communication services
- Current practices of licensing and spectrum assignment in land mobile radio systems particularly in VHF and UHF person-to-person voice communication system in SATRC countries
- Spectrum approaches and regulatory requirements to enable Satellite-IMT integration
- A) Spectrum outlook in SATRC countries, B) Registration of frequency assignments in MIFR by SATRC countries
- Backhaul infrastructure readiness and requirements for 5G deployment in SATRC countries
- Efficient valuation and pricing of spectrum in SATRC countries

(Source: https://www.apt.int)

ABU hosts A26 Global Coordination Conference

The A26 Global Coordination Conference is currently underway in Kuala Lumpur, Malaysia. Approximately fifty frequency managers from across Asia, Australia, the Arab States, Europe, and the Americas are participating in the five-day conference, held from 19 to 23 January 2026.

The Asia-Pacific Broadcasting Union (ABU) is hosting the conference and jointly organised by the High Frequency Coordination Conference (HFCC) and the Arab States Broadcasting Union (ASBU).



The conference opened with a welcome address by ABU Director of Technology, Dr. VeyselBinbay, who extended his warm greetings to all delegates and expressed his hope that the event would be successful and become a memorable experience for all participants.

The conference serves as a key platform for coordinating shortwave frequency schedules for the A26 (summer) season. Shortwave frequency allocations are subject to change due to factors such as the 11-year solar cycle, seasonal variations, and the introduction of new or modified frequencies by broadcasters worldwide. Effective coordination is therefore essential to minimise interference and ensure reliable shortwave broadcasting services.

(Source: https://www.abu.my)

Waves OTT partners with CRIS to bring content into RailOne app

Waves OTT, PrasarBharati's fast-growing digital entertainment platform, has announced a partnership with the Centre for Railway Information Systems (CRIS) to integrate its content library into the RailOne app — the official platform used by millions daily for booking tickets, checking train status, and accessing travel services. Passengers across the country will now be able to watch and listen to Waves content whilst travelling, marking a major expansion for the platform.

Through RailOne, travellers can access live television channels, Akashvani radio stations, movies, documentaries, short films, devotional content, and archival programming in over 15 Indian languages. With nearly 23 million passengers travelling by train each day, this integration significantly extends Waves OTT's reach and strengthens its role as an accessible public service platform.

CEO of PrasarBharati, Gaurav Dwivedi described the partnership as a significant milestone, noting that bringing Waves content to RailOne would greatly enhance passengers' digital experience during travel. The organisations plan a joint promotional campaign to highlight the collaboration and enrich journeys with entertainment services. This initiative reinforces PrasarBharati's commitment to making quality content available anytime and anywhere, now extending to India's vast rail-travelling population.

(Source: https://mib.gov.in)

About The Author



Mr. Rajeev Kumar, a 1994 batch Indian Engineering Services (IES) officer, is Deputy Director General (Engg) at Prasar Bharati. He has vast experience of working in almost all the areas of TV Broadcasting.

Mr Rajeev Kumar is currently responsible for Spectrum Management, ITU-R/ABU Studies and Monitoring of O & M of Akashvani Network in the country. He has contributed at National and International on spectrum, regulatory, standards & training matters towards development/presentation of National viewpoint, standards & policy. He has published many papers on issues related to TV broadcasting, Spectrum Management and Regulatory matters in National and International Journals.

He has a Master of Engineering degree from Indian Institute of Technology (IIT), Roorkee (India). He was also deputed to China for one year (2003-04) to carry out research on Satellite Broadcasting under UNESCO fellowship. He has represented India (& Doordarshan) in various international forums such as Study Group-6/SG-4/SG-5 of ITU-R,

WRCs, APT, AIBD and ABU.

REPORT

Hyderabad Chapter

Dr. Nuli Namassivaya

One Day Seminar on 'AI in Broadcasting – Present & Future'

BES(I), Hyderabad Chapter, in collaboration with the Sri Vasavi Engineering College, Tadepalligudem, Andhra Pradesh, have organised a One-day Seminar entitled 'AI in Broadcasting – Present & Future' on Saturday, 28th December, 2024. Dr.NuliNamassivaya, GC Member, BES(I) was the Convenor, while Dr. E. Kusuma Kumari of the Sri Vasavi Engineering College acted as the Co-convenor. The Chairman and EC members of BES(I), Hyderabad and the President, Principal, Technical Director and Senior faculty of SVEC have graced the occasion. The audience comprised over 200 members – mainly final year and third year under-graduates of SVEC of various branches as well as other Engineering Colleges in and around Tadepalligudem.

During the Inaugural session, Dr.NuliNamassivaya, GC Member, BES(I) spoken about BES(I), HQ activities and the organisation structure of the Seminar. Sri N.V. Ramana, DDG, DD, Hyderabad and Chairman, BES(I), Hyderabad explained about the Broadcast Engineering Society and the pioneering role it is playing in promoting the broadcast engineering profession in the country. Dr. K. Gnaneshwar Rao, Secretary, BES(I), Hyderabad spoke on the theme of the seminar and its relevance in the prevailing technological scenario. Sri C Subba Rao, DDG, DD, Visakhpatnam spoke about the present technologies available in the DD. Dr. GVNSR Ratnakara Rao, Principal, SVEC exhorted the students to embrace the exciting field of Artificial Intelligence and become its active developers rather than just end-users. Sri L. Sreepathi Rao, Treasurer, BES, Hyderabad proposed a Vote of thanks.

Speaking on the topic 'Generative AI in OTT Platforms', Sri N.V. Ramana elaborated on how broadcast engineering has been undergoing changes from time to time in consonance





with technological changes from time to time. Terrestrial broadcasting, as far as Television is concerned, has wound up and OTT (Over The Top) is on the ascendency. With the increasing penetration of AI in broadcasting, OTT is likely to gain further ground and become the favoured mode of distribution in the near future, he said.

'5G & 6G – the India Story' was the topic of the next lecture delivered by Dr. D. Ranganadham, Retd. DDG, DD, Vijayawada, in which the audience were treated to the evolutionary path through which the different generations of data networks have progressed and are progressing, with AI catapulting their phenomenal growth. Dr.NuliNamassivaya, in his lecture 'Influence of AI in Broadcasting' dealt with the basic concepts underlying AI, its application areas and how, broadcasting, in particular, is undergoing radical changes under its influence. The last lecture delivered by Dr.E. Kusuma Kumari on 'Ethics in Broadcasting' gave the audience an overview of the ethical principles governing broadcasting as well as other mass media and how they have become even more pertinent in the unfolding scenario of AI.

Rounding up the day's proceedings, Dr.V. Rama Rao, EC Member, BES, Hyderabad said that AI is a double-edged weapon which has the potential to either elevatehumanity to a higher plane of civilisation or destroy human race altogether. Utmost caution is therefore advisable for future research on this all-powerful and all-pervasive technological megalith.

This seminar is part of the outreach programme taken up by BES(I), Hyderabad through which the Centre is planning to revamp BES activities and bring about awareness among engineering students based in rural /remote areas on broadcast technology and latest developments.

One Day Seminar on 'The Changing Face of Broadcast Technology'

BES, Hyderabad, in collaboration with the GITAM University, Visakhapatnam, A.P. organized a one-day seminar on the theme 'The Changing Face of Broadcast Technology ' on March 7, 2025 at the University premises in Visakhapatnam.Over a hundred B.Tech students, faculty members of the Electronics & Communications department of the University, their senior management and EC members of BES, Hyderabad attended.



At the outset, Prof. Nagendra Prasad, Director, GST of GITAM , in his opening remarks, stated that broadcast technology is advancing by leaps and bounds and this seminar being held at the behest of BES, Hyderabad is an admirable opportunity for the students to keep abreast of the changing technological scenario. Sri K. Subba Rao, DDG, AIR, Hyderabad and Sri L. Sreepathi Rao, Treasurer, BES, Hyderabad briefly explained to the audience the pioneering role BES is playing in the cause on Broadcast Engineering and exhorted them to join the Society as student members so as to benefit from the BES activities.

Dr. D. Ranganadham, former DDG, DDK, Vijayawada, spoke on the theme '5G and Beyond'. He traced the technological evolution of mobile telecommunications over the years and the generational changes that are taking place in this area. He said that the technology is leap-frogging in roughly ten- year intervals, with new value-added services at each step. Now we are at the stage of 5G and soon will be embracing 6G and so on. The increasing data rates, decreasing latency and improved interactivity that these advancements offer us will result in application areas hitherto beyond our reach, he said.



Dr. Nuli Namassivaya, Council Member, BES and Associate Professor, MVSR Engineering College, Hyderabad, spoke on the topic 'Role of AI in Broadcast Technology'. He outlined the fast-emerging field of AI and the different ways in which it is impacting Program production, Post-Production, Storage and Distribution aspects. He also touched upon how the unfolding OTT platforms will transform Broadcast Technology in ways unimagined in the past.

The post-lunch session was handled by Dr. T. Krushna Rao, former Dy.Director, AVPRC, BR Ambedkar Open University, Hyderabad who spoke on ' Digital Audio Broadcasting – Focus on DRM'. He explained the different digital radio broadcast systems in vogue in different countries and their features. Digital Radio Mondiale (DRM), he said, is a dominant digital radio system which is now under experimental stage in our country with several DRM transmitters set up by AIR. Feedback from the test transmissions is very encouraging so far, he said. He also demonstrated some DRM receiver types in use for mobile and portable applications. The seminar was summed up by Dr. V. Rama Rao, EC member, BES, Hyderabad.





Corporate Members

	Mem. No.	Name	Address
1.	LCM-01	M/s AKG Acoustics (India) Ltd.	Kamal Cinema Building, Safdarjung Enclave, New Delhi-110029
2.	LCM-02	M/s Telematics Systems Ltd.	Mount Poonamallee Road, Nandambakkam, Chennai- 600 089
3.	LCM-03	M/s Electronics Corpn.of Tamilnadu Ltd.	LLA Buildings, 735, Anna Salai, Chennai- 600 002
4.	LCM-04	M/s Hinditron Tektronix Inst.Ltd.	No.5, Crescent Road, High Grounds, Bangalore - 560 001
5.	LCM-05	M/s Rohde & Schwarz	Liaison Office India, 244, Okhla Industrial Estate, Phase-III, New Delhi - 110 020
6.	LCM-06	M/s Maharashtra Elects.Corpn.Ltd.	D-26, South Extension, Part-I, New Delhi-110049
7.	LCM-07	M/s Asea Brown Boveri Ltd.	Gurunanak Foundation Building, 15-16, Qutab Institutional Area, New Delhi - 110 067
8.	CM-09	M/s Acoustics Engineers	'VASANT' 1286 A, Shivajinagar, Opp.Balgandharava Rangmandir, Pune - 411 005
9.	LCM-10	M/s Gujarat Commns. & Elect.Ltd.	G.I.D.C Industrial Estate, Makarpura, Vadodara - 390 010
10.	LCM-15	M/s Comcon Industries	S-54, Functional Industrial, Estate for Electronics, Okhla Phase-II, New Delhi - 110 020
11.	LCM-19	M/s Satcom Electronics (P) Ltd.	33, B.R.B. Bose Road, 2nd Floor, Calcutta-700001
12.	LCM-21	M/s Army Headquarter Signal Regiment,	1, Army Headquarter, Signal Regiment, Signals Enclave, New Delhi-110010
13.	LCM-35	M/s Benchmark Microsystems Pvt. Ltd.	No.140,Greater Kailash-I, New Delhi-110048
14.	CM-36	M/s Bharat Electronics Ltd	Jalahalli, Bangalore-560013
15.	CM-25	M/s Visual Tech. India PVT.LTD.	A-1, First floor, lajpat Nagar-II, New Delhi-110024.
16.	LCM-08	M/s Webel Mediatronics Ltd.	P-1, Taratala Road, Calcutta - 700 088
17.	LCM-11	M/s Hindustan Photo Films Co.Ltd.	Indu Nagar, Ooty, Tamilnadu-643006.
18.	LCM-14	M/s Broadcast Equipments(I) P.Ltd.	370-371/2, First Floor, Hospital Road, Jangpura, New Delhi-14
19.	LCM-16	M/s R & S Electronics	A-1-24, Ghanshyam Ind. Estate, Veera Desai Road, Andheri (West), Mumbai-400053
20.	LCM-17	M/s Swati Industries	Z-40, Okhala Industrial Area, Phase-II, New Delhi-110020
21.	LCM-20	M/s Elect. Corpn. of India Ltd.	ECIL Post, Hyderabad-500062
22.	LCM-23	M/s BECIL	B-1, Sector 31, Noida-201301
23.	LCM-24	M/s AVF Distributors (I) PVT. LTD.	208 and 12 jagani Ind. Comp., Near ATI, Chunabhatti, Mumbai-400022.
24.	LCM-26	M/s Cinecita Comoptronics Ind. Pvt. Ltd	Parijat House 3rd floor, 1076, off. pr. E. moses Road, Worli, Mumbai-400018

Corporate Members

	Mem. No.	Name	Address
25.	LCM-27	M/s Studio Systems	Sangeeta Aptts. Bldg. 6-C,Flat No. 5. Ground Floor, Behind Lido Cinema, Juhu Road, Mumbai-400049
26.	LCM-28	M/s Canara Lighting Industries Ltd.	Bata Compound Khopet, Pokhran Road No. 1, Thane-400601
27.	LCM-29	M/s AGIV (India) Pvt. Ltd.	SION Chunabhatti Road, Mumbai-400022
28.	LCM-30	M/s Doordarshan	Doordarshan Bhavan, Copernicus Marg, New Delhi-110001
29.	LCM-31	M/s Silicon Graphics Systems (I)(P)Ltd.	228 Udog Vihar Phase-1, Gurgaon
30.	LCM-32	M/s Essel Shyam Communication Ltd	FC-20, Sector 16 A, Noida-201301
31.	LCM-33	M/s All India Radio	Directorate General, Akashvani Bhavan, Sansad Marg, New Delhi-110001
32.	LCM-34	M/s The IEI Delhi State Centre	Engineers Bhavan, Bahadur Shar Zafar Marg, New Delhi-2
33.	LCM-37	M/s Trans World Radio- South Asia	L-15, Green Park, New Delhi-110016
34.	LCM-38	M/s Kathrein India Pvt. Ltd.	4-B-4. Industry Manor, 3rd Floor, A.S.M. Marg, Prabha Devi, Mumbai-25.
35.	LCM-39	M/s Arraycom (India) Limited	B-13, 13/1 & 14, GIDC, Electronics Estate, Sector 25, Gandhinagar, Gujarat
36.	LCM-40	M/s Sun Broadcast Equipments Pvt. Ltd.	3/31, 3rd Floor, West Patel Nagar, New Delhi-110008
37.	LCM-41	M/s Diksaat Transworld Ltd.	No. 12, Balia Avenue, Luz Mylapore, Chennai-600004
38.	LCM-42	M/s Qualcomm India Pvt. Ltd.	3rd Floor, DLF Centre, Sansad Marg, New Delhi-110001
39.	LCM-43	M/s. Monarch Computers Pvt. Ltd.	Monarch House, Royal Palms, Aarey Milk Colony, Goregaon (E) Mumbai-400065
40.	LCM-44	M/s. Setron India Private Limited	E-2 Greater Kailash Enclave-1 New Delhi-110048
41.	LCM-45	M/s. Global Institute of Technology	ITS-1&2, IT Park, EPIP, Sitapura, Jaipur, Rajasthan-302022
42.	LCM-46	M/s. Institute of Management Studies	A-8B IMS Campus Sector-62 Noida Uttar Pradesh-201303
43.	LCM-47	M/s. A. V. Institute of Technology	Vinayaka Mission University OMR, Vinayaka Nagar, Paiyanoor, Tamilnadu-603104
44.	LCM-48	M/s. Harman International India Pvt. Ltd.	RMZ, Ecospace, 3B, 301, Sarhapura Ring Road, Bengaluru-560103, Karnataka
45.	LCM-49	M/s. Semyung India Enterprises Pvt. Ltd.	Plot No: 131, 1st street, VGP Selva nagar Extn. Velachery- 600 042, Tamilnadu
46.	LCM-50	M/s. Sinclair Broadcast Group	10706, Beaver Dam Road,Hunt Valley, MD 21030, USA 55

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